

1. **Docker installation: Install Docker on your local system (Windows, macOS, or Linux). Verify the installation by running the docker --version and docker info commands.**

Step 1:- Install Docker on your local system

```
root@AvejLanjekar:~# apt install docker.io
docker.io is already the newest version (29.1.3-0ubuntu4.1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 30
root@AvejLanjekar:~#
```

Step 2:- Verify the installation by running the docker --version

```
root@AvejLanjekar:~# docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
root@AvejLanjekar:~#
root@AvejLanjekar:~#
```

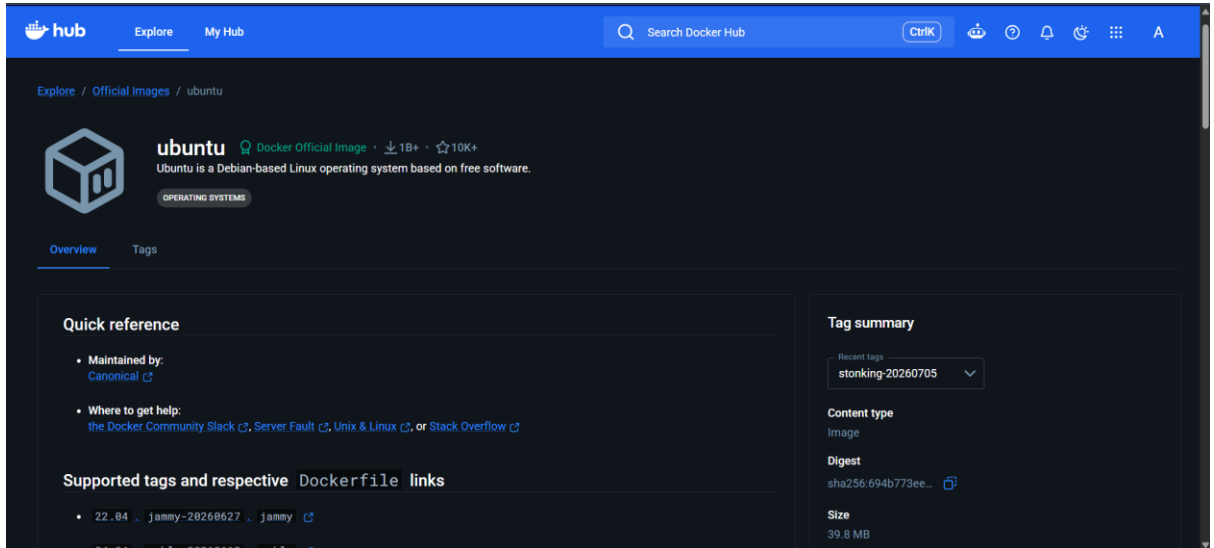
Step 3:- Verify the installation by running docker info

```
root@AvejLanjekar:~# docker info
Client:
Version:      29.1.3
Context:      default
Debug Mode:   false
Plugins:
  compose: Docker Compose (Docker Inc.)
    Version:  2.40.3+ds1-0ubuntu1
    Path:     /usr/libexec/docker/cli-plugins/docker-compose
  trust: Manage trust on Docker images (Docker Inc.)
    Version:  29.1.3
    Path:     /usr/libexec/docker/cli-plugins/docker-trust
Server:
Containers: 1
  Running: 1
  Paused: 0
  Stopped: 0
Images: 8
Server Version: 29.1.3
Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
Logging Driver: json-file
Cgroup Driver: systemd
Cgroup Version: 2
Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
  CDI spec directories:
    /etc/cdi
    /var/run/cdi
Swarm: inactive
Runtimes: io.containerd.runc.v2 runc
Default Runtime: runc
Init Binary: docker-init
containerd version:
runc version:
init version:
Security Options:
  seccomp
    Profile: builtin
  cgroupns
Kernel Version: 6.6.87.2-microsoft-standard-WSL2
Operating System: Ubuntu 26.04 LTS
OSType: linux
Architecture: x86_64
CPUs: 8
Total Memory: 3.725GiB
Name: AvejLanjekar
ID: 300f83c6-2798-44dd-b5c6-d1504f341dfd
Docker Root Dir: /var/lib/docker
Debug Mode: false
Username: avejlanjekar45
Experimental: false
Insecure Registries:
  ::1/128
  127.0.0.0/8
Live Restore Enabled: false
Firewall Backend: iptables

root@AvejLanjekar:~# |
```

2. **Working with Docker images: Search for the official Ubuntu image on Docker Hub, pull it to your local system using docker pull, and list all available images on your system using docker images.**

Step 1 :- Search for the official Ubuntu image on Docker Hub



- Step 2:- pull it to your local system using docker pull and list all available images on your system using docker images.

```
root@AvejLanjekar:~#  
root@AvejLanjekar:~# docker pull ubuntu  
Using default tag: latest  
latest: Pulling from library/ubuntu  
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb  
Status: Image is up to date for ubuntu:latest  
docker.io/library/ubuntu:latest  
root@AvejLanjekar:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	28bd5fe8b56d	13MB	3.93MB	
avejlanjekar45/nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/nginx-img:tagname	5c96dbbd3eef	238MB	63.1MB	
bash:latest	a19c811ee9e9	23.7MB	6.94MB	
centos:7	be65f488b776	301MB	76.1MB	
nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
nginx:latest	5a88c9c45479	241MB	66MB	U
sample-img:latest	fbeb49a8f9fc	238MB	63.1MB	
ubuntu:latest	3131b4cc82a7	161MB	45.3MB	

```
root@AvejLanjekar:~# |
```

3. **Creating a Docker container: Run a Docker container using the Ubuntu image, and execute a command inside the container (e.g., echo "Hello, Docker!").**

```

root@AvejLanjekar:~# docker run -dit --name ubuntu_server ubuntu bash
b79231284a519646497d96d0b774c988f86f1a86d5e1c03d0850ad74c7cf0d41
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                               NAMES
b79231284a51   ubuntu        "/bin/bash"             7 seconds ago   Up 7 seconds   80/tcp                             ubuntu_server
5eecd82dd3f    nginx         "/docker-entrypoint..." 56 minutes ago   Up 56 minutes   80/tcp                             serene_matsumoto
ala34d14cb33   nginx:latest   "/docker-entrypoint..." About an hour ago Up About an hour 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp compose-nginx
root@AvejLanjekar:~#
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker exec -it b79 /bin/bash
root@b79231284a51:/# echo "Hello Docker!"
Hello Docker!
root@b79231284a51:/#

```

4. Inspect the Ubuntu Docker container to ensure it is in the "created" state only

```

root@AvejLanjekar:~# docker container create --name ubuntu_server ubuntu
f8d18ce853398afecba680908280b00b211f7430867975417eab6bf1642fb258
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                               NAMES
5eecd82dd3f    nginx         "/docker-entrypoint..." About an hour ago Up About an hour 80/tcp                             serene_matsumoto
ala34d14cb33   nginx:latest   "/docker-entrypoint..." 2 hours ago      Up 2 hours      0.0.0.0:8080->80/tcp, [::]:8080->80/tcp compose-nginx
root@AvejLanjekar:~# docker container ls -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                               NAMES
f8d18ce85339   ubuntu        "/bin/bash"             15 seconds ago   Created          80/tcp                             ubuntu_server
5eecd82dd3f    nginx         "/docker-entrypoint..." About an hour ago Up About an hour 80/tcp                             serene_matsumoto
ala34d14cb33   nginx:latest   "/docker-entrypoint..." 2 hours ago      Up 2 hours      0.0.0.0:8080->80/tcp, [::]:8080->80/tcp compose-nginx

```

```

root@AvejLanjekar:~# docker container inspect f8d
[
  {
    "Id": "f8d18ce853398afecba680908280b00b211f7430867975417eab6bf1642fb258",
    "Created": "2026-08-03T07:17:47.988789739Z",
    "Path": "/bin/bash",
    "Args": [],
    "State": {
      "Status": "created",
      "Running": false,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 0,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "0001-01-01T00:00:00Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    }
  }
]

```

5. Execute the Ubuntu Docker container and confirm that the container's state is "running."

```

root@AvejLanjekar:~# docker container run -itd --name ubuntu_container ubuntu bash
af3b08b933f5b8c20a49d75b5f32ad24a46373afe0fdbd656cebe01db402da9b
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                               NAMES
af3b08b933f5   ubuntu        "/bin/bash"             11 seconds ago   Up 10 seconds   80/tcp                             ubuntu_container
5eecd82dd3f    nginx         "/docker-entrypoint..." About an hour ago Up About an hour 80/tcp                             serene_matsumoto
ala34d14cb33   nginx:latest   "/docker-entrypoint..." 2 hours ago      Up 2 hours      0.0.0.0:8080->80/tcp, [::]:8080->80/tcp compose-nginx
root@AvejLanjekar:~# docker container inspect af3
[
  {
    "Id": "af3b08b933f5b8c20a49d75b5f32ad24a46373afe0fdbd656cebe01db402da9b",
    "Created": "2026-08-03T07:24:49.807754953Z",
    "Path": "bash",
    "Args": [],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 6560,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-08-03T07:24:49.899426566Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    }
  }
]

```

6. Launch an HTTP container, verify its state, and then remove the container and image from your local system.

Step 1:- Launch an HTTP container, verify its state

```
root@AvejLanjekar:~# docker container run -d --name http-server -P httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
4f4fb780ef54: Pull complete
39d55e966aeb: Pull complete
819f7d7e375f: Pull complete
69b5b831a361: Pull complete
abd81352b960: Pull complete
507729664175: Download complete
c72742279dcl: Download complete
Digest: sha256:305fd8326a27a137fedb8900f26375699ab233cc37793f35cf5d7c65671fb252
Status: Downloaded newer image for httpd:latest
2a528ed9e3c498247ccf5787929ec9823a81c1b25090bb20bc56b5db6f09b5fd
root@AvejLanjekar:~#
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker container inspect 2a52
[
  {
    "Id": "2a528ed9e3c498247ccf5787929ec9823a81c1b25090bb20bc56b5db6f09b5fd",
    "Created": "2026-08-03T07:31:31.105507465Z",
    "Path": "httpd-foreground",
    "Args": [],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 6818,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-08-03T07:31:31.382153662Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    }
  }
],
```

Step 2:- remove the container and image from your local system.

```
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
2a528ed9e3c4   httpd      "httpd-foreground"      About a minute ago    Up 59 seconds    0.0.0.0:32768->80/tcp, [::]:32768->80/tcp    http-server
af3b08b933f5   ubuntu    "bash"                  7 minutes ago        Up 7 minutes     80/tcp                                ubuntu_container
5eecdd82dd3f   nginx     "/docker-entrypoint..." About an hour ago     Up About an hour   80/tcp                                serene_matsumoto
a1a34d14cb33   nginx:late "/docker-entrypoint..." 2 hours ago         Up 2 hours       0.0.0.0:8080->80/tcp, [::]:8080->80/tcp    compose-nginx
root@AvejLanjekar:~#
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker container stop 2a5
2a5
root@AvejLanjekar:~# docker container rm 2a5
2a5
```

```
root@AvejLanjekar:~# docker image rmi httpd
Untagged: httpd:latest
Deleted: sha256:305fd8326a27a137fedb8900f26375699ab233cc37793f35cf5d7c65671fb252
```

7. Execute a command to display a filtered list of containers, showing only their short container IDs.

```
root@AvejLanjekar:~# docker container ls -q
af3b08b933f5
5eecdd82dd3f
a1a34d14cb33
root@AvejLanjekar:~# |
```


8. Clone the lancachenet/ubuntu-nginx Git repository locally and build a Docker image from the provided Dockerfile. Be sure to add relevant tags when building the image.

Step 1:- Clone the lancachenet/ubuntu-nginx Git repository locally

```
root@AvejLanjekar:~#
root@AvejLanjekar:~# git clone https://github.com/lancachenet/ubuntu-nginx.git
Cloning into 'ubuntu-nginx'...
remote: Enumerating objects: 150, done.
remote: Counting objects: 100% (58/58), done.
remote: Compressing objects: 100% (37/37), done.
remote: Total 150 (delta 15), reused 41 (delta 11), pack-reused 92 (from 1)
Receiving objects: 100% (150/150), 33.06 KiB | 3.00 MiB/s, done.
Resolving deltas: 100% (30/30), done.
root@AvejLanjekar:~# ls
compose-demo  docker-compose  docker-project  file.txt  test1  ubuntu-nginx
root@AvejLanjekar:~# cd ubuntu-nginx/
root@AvejLanjekar:~/ubuntu-nginx# ls
Dockerfile  README.md  goss.yaml  goss_wait.yaml  overlay  run-tests.sh
root@AvejLanjekar:~/ubuntu-nginx#
```

Step 2:- build a Docker image from the provided Dockerfile. Be sure to add relevant tags when building the image.

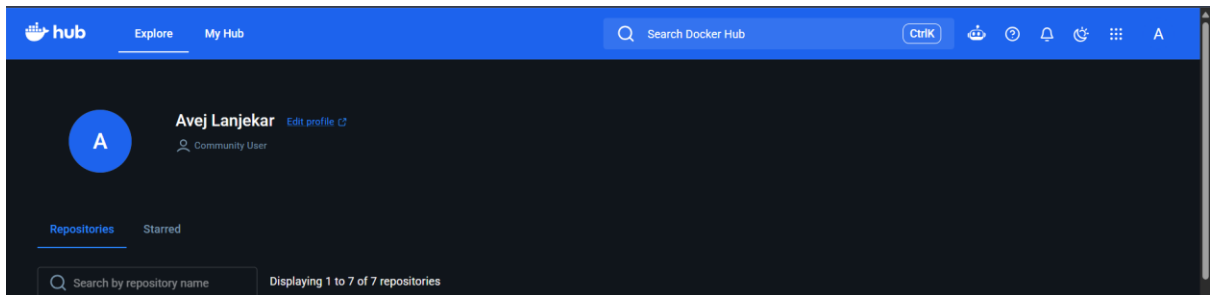
```
root@AvejLanjekar:~/ubuntu-nginx# docker build -t ubutu-nginx-img .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 144.9kB
Step 1/8 : FROM lancachenet/ubuntu:latest
--> f5fde792dd35
Step 2/8 : LABEL org.opencontainers.image.authors="LanCache.Net Team <team@lancache.net>"
--> Using cache
--> 3cd14c8011b7
Step 3/8 : ARG DEBIAN_FRONTEND=noninteractive
--> Using cache
--> f2badcd87451
Step 4/8 : SHELL ["/bin/bash", "-c"]
--> Using cache
--> e07728d20541
Step 5/8 : RUN apt-get update && apt-get install -y nginx-full inotify-tools --no-install-recommends && apt-get clean && rm -rf /var/lib/apt/lists/*
--> Using cache
--> 94a18a622676
Step 6/8 : COPY --link overlay/ /
--> Using cache
--> 6da1371d5b0f
Step 7/8 : RUN rm -f /etc/nginx/sites-available/default /etc/nginx/sites-enabled/default && mkdir -p /etc/nginx/sites-enabled && mkdir -p /etc/nginx/stream-enabled && mkdir -p /var/lib/nginx && mkdir -p /var/www/html && mkdir -p /var/log/nginx && chmod -R 664 /etc/nginx/conf.d/* /etc/nginx/stream.d/* || true && for file in /etc/nginx/sites-available/*; do [ -e "$file" ] && ln -sf "$file" "/etc/nginx/sites-enabled/${basename "$file"}"; done && for file in /etc/nginx/stream-available/*; do [ -e "$file" ] && ln -sf "$file" "/etc/nginx/stream-enabled/${basename "$file"}"; done
--> Using cache
--> 1fa6c1eeaf49
Step 8/8 : EXPOSE 80
--> Using cache
--> aeb87balca5e
Successfully built aeb87balca5e
Successfully tagged ubutu-nginx-img:latest
root@AvejLanjekar:~/ubuntu-nginx#
```

9. After building the image, run a Docker container using the image and check the webpage using the curl command to ensure it's accessible.

```
root@AvejLanjekar:~/ubuntu-nginx# docker container run -d --name webserver -P ubutu-nginx-img:latest
48ba2ff0e8775363faa68694b6cc30060758ed158cdf811076338ebd59fb0c97
root@AvejLanjekar:~/ubuntu-nginx# docker container ls
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAMES
48ba2ff0e877   ubutu-nginx-img:latest   "/bin/bash -e /init/..." 5 seconds ago   Up 5 seconds   0.0.0.0:32774->80/tcp, [::]:32774->80/tcp   webserver
ba2d7bb9114e   ubuntu-nginx:latest     "/bin/bash -e /init/..." 3 minutes ago   Up 3 minutes   0.0.0.0:8084->80/tcp, [::]:8084->80/tcp   ubuntu-nginx-container11
root@AvejLanjekar:~/ubuntu-nginx# curl http://localhost:32774
Welcome to the lancachenet/ubuntu-nginx docker container
root@AvejLanjekar:~/ubuntu-nginx#
```

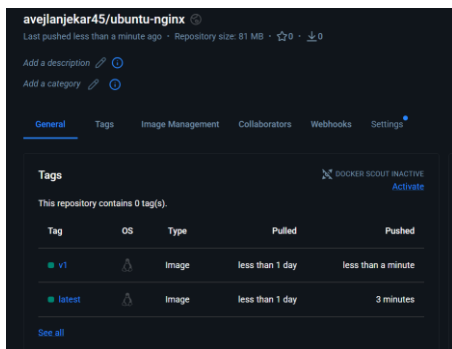
10. Create a DockerHub account, set up a repository, and tag the locally created Docker image using your_repo_name/application_name:version_number.



```
root@AvejLanjekar:~/ubuntu-nginx# docker tag ubutu-nginx-img:latest avejlanjekar45/ubuntu-nginx
root@AvejLanjekar:~/ubuntu-nginx# docker images

IMAGE                                ID                                DISK USAGE    CONTENT SIZE  EXTRA
alpine:latest                        28bd5fe8b56d                      13MB           3.93MB
avejlanjekar45/nginx-img:latest      5c96dbbd3eef                      238MB          63.1MB
avejlanjekar45/nginx-img:tagname     5c96dbbd3eef                      238MB          63.1MB
avejlanjekar45/ubuntu-nginx:latest  aeb87ba1ca5e                      334MB          85MB      U
bash:latest                          a19c811ee9e9                      23.7MB         6.94MB
centos:7                             be65f488b776                      301MB          76.1MB
lancachenet/ubuntu:latest            f5fdc792dd35                      267MB          66MB
nginx-img:latest                     5c96dbbd3eef                      238MB          63.1MB
nginx:latest                         5a88c9c45479                      241MB          66MB
sample-img:latest                   fbeb49a8f9fc                      238MB          63.1MB
ubuntu-nginx1:latest                 76c1ac2b237d                      267MB          66MB
ubuntu-nginx:latest                  aeb87ba1ca5e                      334MB          85MB      U
ubuntu:latest                        3131b4cc82a7                      161MB          45.3MB
ubuntu_nginx:latest                  5649a9b1ea74                      267MB          66MB      U
ubuntu_nginx:latest                  5649a9b1ea74                      267MB          66MB      U
ubutu-nginx-img:latest               aeb87ba1ca5e                      334MB          85MB      U
root@AvejLanjekar:~/ubuntu-nginx# docker push avejlanjekar45/ubuntu-nginx:latest
The push refers to repository [docker.io/avejlanjekar45/ubuntu-nginx]
41140838cd96: Mounted from lancachenet/ubuntu
507c9147a3f2: Pushed
5cf16a7dba6c: Pushed
1eee5a6bd63a: Mounted from lancachenet/ubuntu
093418793dcb: Mounted from lancachenet/ubuntu
5a4c4f8d95f3: Pushed
latest: digest: sha256:aeb87ba1ca5edf1acaabd80555fc06b8124f88b015d544477bc4be19439a7758 size: 1638
```

```
root@AvejLanjekar:~/ubuntu-nginx# docker tag avejlanjekar45/ubuntu-nginx:latest avejlanjekar45/ubuntu-nginx:v1
root@AvejLanjekar:~/ubuntu-nginx# docker push avejlanjekar45/ubuntu-nginx:v1
The push refers to repository [docker.io/avejlanjekar45/ubuntu-nginx]
41140838cd96: Layer already exists
1eee5a6bd63a: Layer already exists
093418793dcb: Layer already exists
5cf16a7dba6c: Layer already exists
5a4c4f8d95f3: Layer already exists
507c9147a3f2: Layer already exists
v1: digest: sha256:aeb87ba1ca5edf1acaabd80555fc06b8124f88b015d544477bc4be19439a7758 size: 1638
root@AvejLanjekar:~/ubuntu-nginx#
```



11. Launch an Ubuntu container, demonstrate that the escape sequence is functional, and reattach the container after detaching it with the escape sequence. Then, check the container's IP address.

```
root@AvejLanjekar:~# docker run -it --name ubuntu-cont ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
a3679419df18: Pull complete
ed819469700f: Pull complete
e16351a257e4: Download complete
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Downloaded newer image for ubuntu:latest
root@33aa32ac58a0:/# root@AvejLanjekar:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
33aa32ac58a0   ubuntu   "/bin/bash"             49 seconds ago Up 47 seconds        0.0.0.0:32774->80/tcp, [::]:32774->80/tcp   ubuntu-cont
48ba2ff0e877   aeb87balca5e   "/bin/bash -e /init/_." 28 minutes ago Up 28 minutes        0.0.0.0:32774->80/tcp, [::]:32774->80/tcp   webserver
ba2d7bb9114e   aeb87balca5e   "/bin/bash -e /init/_." 32 minutes ago Up 31 minutes        0.0.0.0:8084->80/tcp, [::]:8084->80/tcp   ubuntu-nginx-container11
root@AvejLanjekar:~# docker attach ubuntu-cont
root@33aa32ac58a0:/# hostname -i
172.17.0.4
root@33aa32ac58a0:/# |
```

12. Use a Docker command to inspect the hostname and /etc/hosts file of the httpd container.

```
root@AvejLanjekar:~# docker run -d --name http-server -P httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
39d55e96aeb: Pull complete
ebd81352b960: Pull complete
010f7d7e175f: Pull complete
4f4fb70ef54: Pull complete
69b5b831a361: Pull complete
507729664175: Download complete
c72742279dc1: Download complete
Digest: sha256:305fd8326a27a137fedb8900f26375699ab233cc37793f35cf5d7c65671fb252
Status: Downloaded newer image for httpd:latest
69ad9030e1cb3e75c49ede8523d5eff41e524555d704d6efd58aa0793477bd2e
root@AvejLanjekar:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
69ad9030e1cb   httpd     "httpd-foreground"       4 seconds ago Up 3 seconds        0.0.0.0:32775->80/tcp, [::]:32775->80/tcp   http-server
48ba2ff0e877   aeb87balca5e   "/bin/bash -e /init/_." 34 minutes ago Up 34 minutes        0.0.0.0:32774->80/tcp, [::]:32774->80/tcp   webserver
ba2d7bb9114e   aeb87balca5e   "/bin/bash -e /init/_." 37 minutes ago Up 37 minutes        0.0.0.0:8084->80/tcp, [::]:8084->80/tcp   ubuntu-nginx-container11
root@AvejLanjekar:~# docker inspect http-server
[
  {
    "Id": "69ad9030e1cb3e75c49ede8523d5eff41e524555d704d6efd58aa0793477bd2e",
    "Created": "2026-08-03T11:12:03.815121519Z",
    "Path": "httpd-foreground",
    "Args": [],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 18260,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-08-03T11:12:04.068652619Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Mounts": [],
    "Config": {
      "Hostname": "69ad9030e1cb",
      "Domainname": "",
      "User": "",
      "AttachStdin": false,
      "AttachStdout": false,
      "AttachStderr": false,
      "ExposedPorts": {
        "80/tcp": {}
      },
      "Tty": false,
      "OpenStdin": false,
      "StdinOnce": false,
      "Env": [
        "PATH=/usr/local/apache2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "HTTPD_PREFIX=/usr/local/apache2",
        "HTTPD_VERSION=2.4.68",
        "HTTPD_SHA256=68c74d4df38c26bed4dfbdb8f3baf1eb532f3872357becc1bba5d136f6b63c06",
        "HTTPD_PATCHES="
      ],
      "Cmd": [
        "httpd-foreground"
      ],
      "Image": "httpd",
      "Volumes": null,
      "WorkingDir": "/usr/local/apache2",
      "Entrypoint": null,
      "Labels": {},
      "StopSignal": "SIGWINCH"
    },
    "NetworkSettings": {
```

```
root@AvejLanjekar:~# docker exec http-server hostname
69ad9030e1cb
root@AvejLanjekar:~# |
```

```

root@AvejLanjekar:~# docker exec http-server cat /etc/hosts
127.0.0.1        localhost
::1             localhost ip6-localhost ip6-loopback
fe00::          ip6-localnet
ff00::          ip6-mcastprefix
ff02:::1        ip6-allnodes
ff02:::2        ip6-allrouters
172.17.0.4      69ad9030e1cb
root@AvejLanjekar:~# |

```

```

root@AvejLanjekar:~# docker exec -it http-server /bin/bash
root@69ad9030e1cb:/usr/local/apache2# ls
bin build cgi-bin conf error htdocs icons include logs modules
root@69ad9030e1cb:/usr/local/apache2# cat /etc/hosts
127.0.0.1        localhost
::1             localhost ip6-localhost ip6-loopback
fe00::          ip6-localnet
ff00::          ip6-mcastprefix
ff02:::1        ip6-allnodes
ff02:::2        ip6-allrouters
172.17.0.4      69ad9030e1cb
root@69ad9030e1cb:/usr/local/apache2# |

```

13. Run a Docker command to inspect a container created with the lancachenet/ubuntu-nginx repository, and monitor the container's resource usage and stats with the TOP command.

```

root@AvejLanjekar:~/ubuntu-nginx# docker inspect ubuntu-server
[
  {
    "Id": "9eabb293d394ecc0a23e60e5d8cabf3d4d8789bc304007e7e10f5a563bbbc33c",
    "Created": "2026-08-03T11:25:19.217450457Z",
    "Path": "/bin/bash",
    "Args": [
      "-e",
      "/init/entrypoint.sh",
      "/init/supervisord.sh"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 19416,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-08-03T11:25:19.303995608Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:aeb87balca5edf1acaabd80555fc06b8124f88b015d544477bc4be19439a7758",
    "ResolvConfPath": "/var/lib/docker/containers/9eabb293d394ecc0a23e60e5d8cabf3d4d8789bc304007e7e10f5a563bbbc33c/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/9eabb293d394ecc0a23e60e5d8cabf3d4d8789bc304007e7e10f5a563bbbc33c/hostname",
    "HostsPath": "/var/lib/docker/containers/9eabb293d394ecc0a23e60e5d8cabf3d4d8789bc304007e7e10f5a563bbbc33c/hosts",
    "LogPath": "/var/lib/docker/containers/9eabb293d394ecc0a23e60e5d8cabf3d4d8789bc304007e7e10f5a563bbbc33c/9eabb293d394ecc0a23e60e5d8cabf3d4d8789bc304007e7e10f5a563bbbc33c-json.log",
    "Name": "/ubuntu-server",
    "RestartCount": 0,
    "Driver": "overlayfs",
    "Platform": "linux",
    "MountLabel": "",
    "ProcessLabel": "",
    "AppArmorProfile": "",
    "ExecIDs": null,
    "HostConfig": {
      "Binds": null,
      "ContainerIDFile": ""
    }
  }
]

```

```

root@AvejLanjekar:~/ubuntu-nginx# docker top ubuntu-server
root
UID          PID         PPID        C          STATE      TTY          TIME          CMD
root         19416       12394       0          MARB      ?            00:00:00      /usr/bin/python3
root         19467       19416       0          ?         ?            00:00:00      /bin/bash /opt/n
nginx/startng 19468       19467       0          ?         ?            00:00:00      nginx: master pr
cess /usr/sbin/nginx -g daemon off;
www-data     19469       19468       0          ?         ?            00:00:00      nginx: worker pr
cess
www-data     19470       19468       0          ?         ?            00:00:00      nginx: worker pr
cess
www-data     19471       19468       0          ?         ?            00:00:00      nginx: worker pr
cess
www-data     19472       19468       0          ?         ?            00:00:00      nginx: worker pr
cess
www-data     19473       19468       0          ?         ?            00:00:00      nginx: worker pr
cess
www-data     19474       19468       0          ?         ?            00:00:00      nginx: worker pr
cess
www-data     19475       19468       0          ?         ?            00:00:00      nginx: worker pr
cess
www-data     19476       19468       0          ?         ?            00:00:00      nginx: worker pr
cess
root@AvejLanjekar:~/ubuntu-nginx# |

```

```

root@AvejLanjekar:~/ubuntu-nginx# docker stats ubuntu-server |

```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
9eabb293d394	ubuntu-server	0.02%	25.82MiB / 3.725GiB	0.68%	936B / 126B	9.51MB / 24.6kB	11

14. Start a new Nginx container and review the logs for both Nginx containers.

```
root@AvejLanjekar:~/ubuntu-nginx# docker run -d -P --name nginx-cont2 nginx
ff9d2729ec095e671b5b05955d182045522dc8d68b8717b3f35c89d1162cf336
```

```
root@AvejLanjekar:~/ubuntu-nginx# docker logs nginx-cont1
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/08/03 11:37:53 [notice] 1#1: using the "epoll" event method
2026/08/03 11:37:53 [notice] 1#1: nginx/1.31.3
2026/08/03 11:37:53 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/08/03 11:37:53 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/08/03 11:37:53 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026/08/03 11:37:53 [notice] 1#1: start worker processes
2026/08/03 11:37:53 [notice] 1#1: start worker process 29
2026/08/03 11:37:53 [notice] 1#1: start worker process 30
2026/08/03 11:37:53 [notice] 1#1: start worker process 31
2026/08/03 11:37:53 [notice] 1#1: start worker process 32
2026/08/03 11:37:53 [notice] 1#1: start worker process 33
2026/08/03 11:37:53 [notice] 1#1: start worker process 34
2026/08/03 11:37:53 [notice] 1#1: start worker process 35
2026/08/03 11:37:53 [notice] 1#1: start worker process 36
```

```
root@AvejLanjekar:~/ubuntu-nginx# docker logs nginx-cont2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/08/03 11:37:58 [notice] 1#1: using the "epoll" event method
2026/08/03 11:37:58 [notice] 1#1: nginx/1.31.3
2026/08/03 11:37:58 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/08/03 11:37:58 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/08/03 11:37:58 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026/08/03 11:37:58 [notice] 1#1: start worker processes
2026/08/03 11:37:58 [notice] 1#1: start worker process 29
2026/08/03 11:37:58 [notice] 1#1: start worker process 30
2026/08/03 11:37:58 [notice] 1#1: start worker process 31
2026/08/03 11:37:58 [notice] 1#1: start worker process 32
2026/08/03 11:37:58 [notice] 1#1: start worker process 33
2026/08/03 11:37:58 [notice] 1#1: start worker process 34
2026/08/03 11:37:58 [notice] 1#1: start worker process 35
2026/08/03 11:37:58 [notice] 1#1: start worker process 36
root@AvejLanjekar:~/ubuntu-nginx#
```

15. Examine system events, filtering them by date and the last 30 minutes. Additionally, apply two filters simultaneously using name and event.

```
root@AveJlanjekar:~# docker events --since "2026-08-03T00:00:00"
2026-08-03T09:40:23.732632194Z container destroy 514868e238edbebe16001ef2e8d6e81387affee7bca6fcec8fb2519d9b04c6c03 (image=ubuntu-nginx:latest, name=ubuntu-nginx-container, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T09:40:23.779975601Z container destroy 5569ed26e647af4d5eb5dcd39c8e33e5bb9c529166a0d9afc1b4a575af4d251 (image=ubuntu-nginx:latest, name=webpage-server, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T09:40:23.814115840Z container destroy 15e726e872391842743f44eaf5bf477d5b82fald7de79b08d26175e286c12a9 (image=ubuntu-nginx:latest, name=webpage-server, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T09:40:23.845457297Z container destroy ed4c3dacedd619d68f91a33b0cc160de79c58e8278045539db0f54d08027aee25 (image=ubuntu-nginx:latest, name=webpage, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T09:40:23.845469294Z container prune (reclaimed=163840)
2026-08-03T09:41:14.805637370Z image tag sha256:5649a9b1ea7483b1e5121c6e9bc64d526a2b6a12e1ecadd3ec43eb4f4dc71d994 (name=ubuntu-nginx:latest, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T09:41:27.726986825Z container create a14acb5dd6cf53c4c53b137048fcd40ee51f5b33d525e8c98efbc971aea4a6fc (image=ubuntu-nginx:latest, name=ubuntu-nginx-container, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T09:41:27.959585818Z network connect 6a713c7b859e50fbc9cf2d2c11bf4985435710967374e07d472a1b1abb5b85f (container=a14acb5dd6cf53c4c53b137048fcd40ee51f5b33d525e8c98efbc971aea4a6fc, name=bridge, type=bridge)
2026-08-03T09:41:27.978204767Z container start a14acb5dd6cf53c4c53b137048fcd40ee51f5b33d525e8c98efbc971aea4a6fc (image=ubuntu-nginx:latest, name=ubuntu-nginx-container, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T09:41:28.297124584Z network disconnect 6a713c7b859e50fbc9cf2d2c11bf4985435710967374e07d472a1b1abb5b85f (container=a14acb5dd6cf53c4c53b137048fcd40ee51f5b33d525e8c98efbc971aea4a6fc, name=bridge, type=bridge)
2026-08-03T09:41:28.327634562Z container die a14acb5dd6cf53c4c53b137048fcd40ee51f5b33d525e8c98efbc971aea4a6fc (execDuration=0, exitCode=1, image=ubuntu-nginx:latest, name=ubuntu-nginx-container, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T09:45:01.073491145Z container kill 5eecdd82dd3f56e7add1bf787753b44658a1ec7b3e6397972b34d2531dae1627 (image=nginx, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=serene_matsumoto, signal=9)
2026-08-03T09:45:01.39669620Z network disconnect 6a713c7b859e50fbc9cf2d2c11bf4985435710967374e07d472a1b1abb5b85f (container=5eecdd82dd3f56e7add1bf787753b44658a1ec7b3e6397972b34d2531dae1627, name=bridge, type=bridge)
2026-08-03T09:45:01.411892980Z container die 5eecdd82dd3f56e7add1bf787753b44658a1ec7b3e6397972b34d2531dae1627 (execDuration=12784, exitCode=137, image=nginx, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=serene_matsumoto)
2026-08-03T09:45:01.440296572Z container destroy 5eecdd82dd3f56e7add1bf787753b44658a1ec7b3e6397972b34d2531dae1627 (image=nginx, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=serene_matsumoto)
2026-08-03T09:45:08.767982827Z container kill a1a34d14cb337811a2bd07770e1a80b386173a80ed45a35ff62b6600be980f70 (com.docker.compose.config-hash=c7756e5151f2375884aa6e04f4f0661128913d7239a5b2208e580cd7c3787d63, com.docker.compose.container-number=1, com.docker.compose.depends_on, com.docker.compose.image=sha256:5a8bc9c7499443d7b2e2adcb89b4ed0a9801bae03d97a5760ae13b5c2005942, com.docker.compose.oneoff=False, com.docker.compose.project=nginx, com.docker.compose.project.config-files=/root/docker-compose/docker-compose.yml, com.docker.compose.project.working_dir=/root/docker-compose, com.docker.compose.service=nginx, com.docker.compose.version=2.40.3, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=compose-nginx, signal=9)
2026-08-03T09:45:09.269246245Z network disconnect 18f0646935f9a5e18942c8c0c7a8d33b54184dc6e8261e881ac9ac8b699ca6747 (container=a1a34d14cb337811a2bd07770e1a80b386173a80ed45a35ff62b6600be980f70, driver=local)
2026-08-03T09:45:09.286488514Z volume unmount docker-compose-nginx-data (container=a1a34d14cb337811a2bd07770e1a80b386173a80ed45a35ff62b6600be980f70, driver=local)
2026-08-03T09:45:09.29357774Z container die a1a34d14cb337811a2bd07770e1a80b386173a80ed45a35ff62b6600be980f70 (com.docker.compose.config-hash=c7756e5151f237
```

```
root@AveJlanjekar:~# docker system events --since 30m
2026-08-03T11:19:17.564188970Z container exec.create: cat /etc/hosts 69ad9030e1cb3e75c49ede8523d5eff41e524555d704d6efd58aa0793477bd2e (execID=187ca7bb28baa2d8fde7a8f7b6b288e4d712aa19e8545b979e19831c4d435b1, image=httpd, name=http-server)
2026-08-03T11:19:17.568392966Z container exec.start: cat /etc/hosts 69ad9030e1cb3e75c49ede8523d5eff41e524555d704d6efd58aa0793477bd2e (execID=187ca7bb28baa2d8fde7a8f7b6b288e4d712aa19e8545b979e19831c4d435b1, image=httpd, name=http-server)
2026-08-03T11:19:17.602827516Z container exec.die 69ad9030e1cb3e75c49ede8523d5eff41e524555d704d6efd58aa0793477bd2e (execID=187ca7bb28baa2d8fde7a8f7b6b288e4d712aa19e8545b979e19831c4d435b1, exitCode=0, image=httpd, name=http-server)
2026-08-03T11:20:51.4433999946Z container exec.create: /bin/bash 69ad9030e1cb3e75c49ede8523d5eff41e524555d704d6efd58aa0793477bd2e (execID=43e7ec4e08c508e92b816efe3429e6940d1dcb8ecd09986346fc11b21a23ba12d, image=httpd, name=http-server)
2026-08-03T11:20:51.444506570Z container exec.start: /bin/bash 69ad9030e1cb3e75c49ede8523d5eff41e524555d704d6efd58aa0793477bd2e (execID=43e7ec4e08c508e92b816efe3429e6940d1dcb8ecd09986346fc11b21a23ba12d, image=httpd, name=http-server)
2026-08-03T11:22:11.566318670Z container exec.die 69ad9030e1cb3e75c49ede8523d5eff41e524555d704d6efd58aa0793477bd2e (execID=43e7ec4e08c508e92b816efe3429e6940d1dcb8ecd09986346fc11b21a23ba12d, exitCode=0, image=httpd, name=http-server)
2026-08-03T11:24:38.961386986Z image pull lancahenet/ubuntu:latest (name=Lancahenet/ubuntu, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T11:24:39.063662017Z image tag sha256:aeb87ba1ca5edf1acaabd08555fc6b68124f88b015d544477bc4be19439a7758 (name=ubuntu-nginx:latest, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T11:25:19.277978874Z container create 9eabb293d394ecc0a23e60e5d8cabf3d4d8789bc304007e7e10f5a563bbbc33c (image=ubuntu-nginx:latest, name=ubuntu-server, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T11:25:19.508984389Z network connect 6a713c7b859e50fbc9cf2d2c11bf4985435710967374e07d472a1b1abb5b85f (container=9eabb293d394ecc0a23e60e5d8cabf3d4d8789bc304007e7e10f5a563bbbc33c, name=bridge, type=bridge)
2026-08-03T11:25:19.515597641Z container start 9eabb293d394ecc0a23e60e5d8cabf3d4d8789bc304007e7e10f5a563bbbc33c (image=ubuntu-nginx:latest, name=ubuntu-server, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T11:29:45.380488869Z container top 69ad9030e1cb3e75c49ede8523d5eff41e524555d704d6efd58aa0793477bd2e (image=httpd, name=http-server)
2026-08-03T11:30:22.295335380Z container top 9eabb293d394ecc0a23e60e5d8cabf3d4d8789bc304007e7e10f5a563bbbc33c (image=ubuntu-nginx:latest, name=ubuntu-server, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
2026-08-03T11:37:52.716709863Z container create f1d06c93f50bbf38353811c0ad1e7e0040d8ead02382f06f1ba048050e2ced88 (image=nginx, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=nginx-cont1)
2026-08-03T11:37:53.059491988Z network connect 6a713c7b859e50fbc9cf2d2c11bf4985435710967374e07d472a1b1abb5b85f (container=f1d06c93f50bbf38353811c0ad1e7e0040d8ead02382f06f1ba048050e2ced88, name=bridge, type=bridge)
2026-08-03T11:37:53.078936120Z container start f1d06c93f50bbf38353811c0ad1e7e0040d8ead02382f06f1ba048050e2ced88 (image=nginx, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=nginx-cont1)
2026-08-03T11:37:57.870870523Z container create ff9d2729ec095e671b5b05955d182045522cd8d68b8717b3f35c89d1162cf336 (image=nginx, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=nginx-cont2)
2026-08-03T11:37:58.002015931Z network connect 6a713c7b859e50fbc9cf2d2c11bf4985435710967374e07d472a1b1abb5b85f (container=ff9d2729ec095e671b5b05955d182045522cd8d68b8717b3f35c89d1162cf336, name=bridge, type=bridge)
2026-08-03T11:37:58.097117606Z container start ff9d2729ec095e671b5b05955d182045522cd8d68b8717b3f35c89d1162cf336 (image=nginx, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=nginx-cont2)
```

```
root@AveJlanjekar:~# docker events --filter container=ubuntu-server --filter event=start
2026-08-03T11:54:06.520236682Z container start 9eabb293d394ecc0a23e60e5d8cabf3d4d8789bc304007e7e10f5a563bbbc33c (image=ubuntu-nginx:latest, name=ubuntu-server, org.opencontainers.image.authors=LanCache.Net Team <team@lancache.net>, org.opencontainers.image.version=24.04)
```

16. Create a Docker container using the Ubuntu image, assign it a meaningful name, and set the hostname to "cloudeithx.com." Stop or kill the container, check its exit code, and examine the container's log for more information.

```
root@AvejLanjekar:~# docker run -itd --name cloudeithx-ubuntu-server --hostname cloudeithx.com ubuntu
12a0a0761135b94cf7f24d45c8e05ac6677d5f8b3e408ffc2413cb7947acc76a7
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
12a0a0761135   ubuntu        "/bin/bash"             19 seconds ago Up 18 seconds                               cloudeithx-
ubuntu-server
ff9d2729ec09   nginx         "/docker-entrypoint..." 21 minutes ago Up 21 minutes   0.0.0.0:32777->80/tcp, [::]:32777->80/tcp   nginx-cont2
f1d06c93f50b   nginx         "/docker-entrypoint..." 22 minutes ago Up 22 minutes   0.0.0.0:32776->80/tcp, [::]:32776->80/tcp   nginx-cont1
9eabb293d394   ubuntu-nginx:latest "/bin/bash -e /init/..." 34 minutes ago Up 5 minutes    80/tcp                                     ubuntu-serv
er
69ad9030e1cb   httpd         "httpd-foreground"       47 minutes ago Up 6 minutes    0.0.0.0:32778->80/tcp, [::]:32778->80/tcp   http-server
48ba2ff0e877   aeb87balca5e "/bin/bash -e /init/..." About an hour ago Up About an hour 0.0.0.0:32774->80/tcp, [::]:32774->80/tcp   webserver
ba2d7bb9114e   ubuntu-nginx:latest "/bin/bash -e /init/..." About an hour ago Up About an hour 0.0.0.0:8084->80/tcp, [::]:8084->80/tcp   ubuntu-ngin
x-container11
root@AvejLanjekar:~# docker exec -it cloudeithx-ubuntu-server /bin/bash
root@cloudeithx:~# hostname
cloudeithx.com
root@cloudeithx:~# |

root@AvejLanjekar:~#
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker inspect cloudeithx-ubuntu-server --format='{{.State.ExitCode}}'
137
root@AvejLanjekar:~# docker logs cloudeithx-ubuntu-server
root@cloudeithx:~# root@AvejLanjekar:~# |
```

17. Instantiate and start an Ubuntu container with a random name, then rename it to "Ubuntu01." Launch another container named "Ubuntu02" and check the hostname of both containers. Pause and un-pause the "Ubuntu02" container, and stop, start, and restart the "Ubuntu01" container. Inspect the stats and system events of both containers, then kill and delete them, ensuring that directories created by the containers are removed.

Step 1:- Instantiate and start an Ubuntu container with a random name

```
root@AvejLanjekar:~# docker run -itd ubuntu
17b5f4dbdcf3f3d65594f8d527020971a36fed7ae364c07db667e4673f14d5a5
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
17b5f4dbdcf3   ubuntu        "/bin/bash"             18 seconds ago Up 17 seconds                               unruffled_cannon
```

Step 2:- rename it to "Ubuntu01."

```
root@AvejLanjekar:~# docker container rename unruffled_cannon ubuntu_01
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
17b5f4dbdcf3   ubuntu        "/bin/bash"             47 seconds ago Up 47 seconds                               ubuntu_01
ff9d2729ec09   nginx         "/docker-entrypoint..." 32 minutes ago Up 32 minutes   0.0.0.0:32777->80/tcp, [::]:32777->80/tcp   nginx-cont2
f1d06c93f50b   nginx         "/docker-entrypoint..." 32 minutes ago Up 32 minutes   0.0.0.0:32776->80/tcp, [::]:32776->80/tcp   nginx-cont1
```

Step 3:- Launch another container named "Ubuntu02"

```
root@AvejLanjekar:~# docker run -itd --name ubuntu02 ubuntu
194d12958d66848338c13c585398aa0437fb80e9edc13adcdecdf92f415c1909
root@AvejLanjekar:~# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
194d12958d66   ubuntu        "/bin/bash"             3 seconds ago Up 2 seconds                               ubuntu02
17b5f4dbdcf3   ubuntu        "/bin/bash"             2 minutes ago Up 2 minutes                               ubuntu_01
ff9d2729ec09   nginx         "/docker-entrypoint..." 34 minutes ago Up 34 minutes   0.0.0.0:32777->80/tcp, [::]:32777->80/tcp   nginx-cont2
f1d06c93f50b   nginx         "/docker-entrypoint..." 34 minutes ago Up 34 minutes   0.0.0.0:32776->80/tcp, [::]:32776->80/tcp   nginx-cont1
9eabb293d394   ubuntu-nginx:latest "/bin/bash -e /init/..." 47 minutes ago Up 18 minutes    80/tcp                                     ubuntu-server
69ad9030e1cb   httpd         "httpd-foreground"       About an hour ago Up 19 minutes    0.0.0.0:32778->80/tcp, [::]:32778->80/tcp   http-server
48ba2ff0e877   aeb87balca5e "/bin/bash -e /init/..." 2 hours ago Up 2 hours      0.0.0.0:32774->80/tcp, [::]:32774->80/tcp   webserver
ba2d7bb9114e   ubuntu-nginx:latest "/bin/bash -e /init/..." 2 hours ago Up 2 hours      0.0.0.0:8084->80/tcp, [::]:8084->80/tcp   ubuntu-nginx-c
ontainer11
root@AvejLanjekar:~# |
```

Step 4:- check the hostname of both containers

```

root@AvejLanjekar:~# docker exec ubuntu_01 hostname
17b5f4dbdcf3
root@AvejLanjekar:~# docker exec ubuntu02 hostname
194d12958d66
root@AvejLanjekar:~# |

```

Step 5:- Pause and un-pause the "Ubuntu02" container

```

root@AvejLanjekar:~# docker container pause ubuntu02
ubuntu02
root@AvejLanjekar:~# docker container unpause ubuntu02
ubuntu02
root@AvejLanjekar:~#

```

Step 6:- stop, start, and restart the "Ubuntu01" container

```

root@AvejLanjekar:~# docker container stop ubuntu_01
ubuntu_01
root@AvejLanjekar:~# docker container start ubuntu_01
ubuntu_01
root@AvejLanjekar:~# docker container restart ubuntu_01
ubuntu_01
root@AvejLanjekar:~# |

```

Step 7:- Inspect the stats and system events of both containers

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
17b5f4dbdcf3	ubuntu_01	0.00%	888KiB / 3.725GiB	0.02%	796B / 126B	0B / 0B	1
194d12958d66	ubuntu02	0.00%	948KiB / 3.725GiB	0.02%	1.26kB / 126B	0B / 0B	1

On one terminal run cmd:- docker system events --filter container=ubuntu_01
&

On another terminal run some commands on ubuntu_01 , here we have stop
ubutu_01 cont and see the result of events on terminal 1

```

root@AvejLanjekar:~# docker container stop ubuntu_01
ubuntu_01
root@AvejLanjekar:~# |

```

```

root@AvejLanjekar:~#
root@AvejLanjekar:~# docker system events --filter container=ubuntu_01
2026-08-03T12:28:42.646485466Z container kill 17b5f4dbdcf3f3d65594f8d527020971a36fed7ae364c07db667e4673f14d5a5 (image=ubuntu, name=ubuntu_01, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

Ubuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
It is the world's most popular operating system across public clouds and OpenStack clouds.
It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04, signal=15)
2026-08-03T12:28:52.668562210Z container kill 17b5f4dbdcf3f3d65594f8d527020971a36fed7ae364c07db667e4673f14d5a5 (image=ubuntu, name=ubuntu_01, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

Ubuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
It is the world's most popular operating system across public clouds and OpenStack clouds.
It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04, signal=9)
2026-08-03T12:28:52.875417946Z container stop 17b5f4dbdcf3f3d65594f8d527020971a36fed7ae364c07db667e4673f14d5a5 (image=ubuntu, name=ubuntu_01, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

Ubuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
It is the world's most popular operating system across public clouds and OpenStack clouds.
It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04)
2026-08-03T12:28:52.880014931Z container die 17b5f4dbdcf3f3d65594f8d527020971a36fed7ae364c07db667e4673f14d5a5 (execDuration=315, exitCode=137, image=ubuntu, name=ubuntu_01, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

Ubuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
It is the world's most popular operating system across public clouds and OpenStack clouds.
It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04)

```


Kill the cont ubuntu02

```
root@AvejLanjekar:~# docker container kill ubuntu02
ubuntu02
```

Delete the cont ubuntu02

```
root@AvejLanjekar:~# docker container rm ubuntu02
ubuntu02
root@AvejLanjekar:~# |
```

18. Run four HTTPD Docker containers with distinct, meaningful names, and apply restart policies (NO, On-Failure, Always, and Unless-Stopped) to each of the four containers, respectively. Demonstrate that the restart policies function as expected.

Step 1:- Container 1 – No Restart Policy

```
root@AvejLanjekar:~# docker run -d --name httpd-no --restart no httpd
82b7ef5d2e8ff3bb77433c94ff9d19fef55d61190f7dec11759c858f830276a1
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
82b7ef5d2e8f   httpd    "httpd-foreground"      9 seconds ago Up 8 seconds  80/tcp       httpd-no
root@AvejLanjekar:~# |
```

Step 2:- Container 2 – On Failure

```
root@AvejLanjekar:~# docker run -d --name httpd-failure --restart on-failure httpd
3b1987d5c0e834f93a6db1a61b728f625371e7fc4b597211352c6c887757ab6f
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
3b1987d5c0e8   httpd    "httpd-foreground"      3 seconds ago Up 2 seconds  80/tcp       httpd-failure
82b7ef5d2e8f   httpd    "httpd-foreground"      About a minute ago Up About a minute  80/tcp       httpd-no
root@AvejLanjekar:~# |
```

Step 3: Container 3 – Always

```
root@AvejLanjekar:~# docker run -d --name httpd-always --restart always httpd
e0ed2dd929db35fcc8b09a5c17a10357149b929dc1999b8d9b9e8fae6146e8b
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
e0ed2dd929db   httpd    "httpd-foreground"      3 seconds ago Up 2 seconds  80/tcp       httpd-always
3b1987d5c0e8   httpd    "httpd-foreground"      About a minute ago Up About a minute  80/tcp       httpd-failure
82b7ef5d2e8f   httpd    "httpd-foreground"      2 minutes ago Up 2 minutes  80/tcp       httpd-no
root@AvejLanjekar:~# |
```

Step 4:- Container 4 – Unless Stopped

```
root@AvejLanjekar:~# docker run -d --name httpd-unless --restart unless-stopped httpd
921393580e188a9c47beba575aefb4266f8353b4ae2cb65984e135bc2103811
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
921393580e18   httpd    "httpd-foreground"      2 seconds ago Up 2 seconds  80/tcp       httpd-unless
e0ed2dd929db   httpd    "httpd-foreground"      About a minute ago Up About a minute  80/tcp       httpd-always
3b1987d5c0e8   httpd    "httpd-foreground"      2 minutes ago Up 2 minutes  80/tcp       httpd-failure
82b7ef5d2e8f   httpd    "httpd-foreground"      4 minutes ago Up 4 minutes  80/tcp       httpd-no
root@AvejLanjekar:~# |
```

Demonstrate that the restart policies function as expected.

Restart Policy 1:- NO

```
root@AvejLanjekar:~# docker stop httpd-no
httpd-no
root@AvejLanjekar:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS      NAMES
921393580e18   httpd     "httpd-foreground"      About a minute ago    Up About a minute    80/tcp     httpd-unless
e0ed2dd929db   httpd     "httpd-foreground"      3 minutes ago        Up 2 minutes        80/tcp     httpd-always
3b1987d5c0e8   httpd     "httpd-foreground"      4 minutes ago        Up 4 minutes        80/tcp     httpd-failure
root@AvejLanjekar:~# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS      NAMES
921393580e18   httpd     "httpd-foreground"      About a minute ago    Up About a minute    80/tcp     httpd-unless
e0ed2dd929db   httpd     "httpd-foreground"      3 minutes ago        Up 3 minutes        80/tcp     httpd-always
3b1987d5c0e8   httpd     "httpd-foreground"      4 minutes ago        Up 4 minutes        80/tcp     httpd-failure
82b7ef5d2e8f   httpd     "httpd-foreground"      5 minutes ago        Exited (0) 8 seconds ago              httpd-no
root@AvejLanjekar:~#
```

Docker does nothing. Container stays stopped.

Restart Policy 2:- On Failure

```
root@AvejLanjekar:~# docker stop httpd-failure
httpd-failure
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS      NAMES
921393580e18   httpd     "httpd-foreground"      9 minutes ago    Up 9 minutes    80/tcp     httpd-unless
e0ed2dd929db   httpd     "httpd-foreground"      10 minutes ago   Up 10 minutes    80/tcp     httpd-always
root@AvejLanjekar:~# docker container ls -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS      NAMES
921393580e18   httpd     "httpd-foreground"      9 minutes ago    Up 9 minutes    80/tcp     httpd-unless
e0ed2dd929db   httpd     "httpd-foreground"      11 minutes ago   Up 11 minutes    80/tcp     httpd-always
3b1987d5c0e8   httpd     "httpd-foreground"      12 minutes ago   Exited (0) 15 seconds ago              httpd-failure
82b7ef5d2e8f   httpd     "httpd-foreground"      13 minutes ago   Exited (0) 8 minutes ago              httpd-no
root@AvejLanjekar:~#
```

This policy only restarts the container if the main process exits with a non-zero exit code (an error). In our example which we are stopping cont will not restart it, because docker stop sends a graceful stop signal, resulting in exit code 0.

Restart Policy 3:- always

The always policy causes Docker to restart the container automatically after the Docker daemon starts again.

```
root@AvejLanjekar:~# docker container stop httpd-always
httpd-always
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS      NAMES
921393580e18   httpd     "httpd-foreground"      17 minutes ago    Up About a minute    80/tcp     httpd-unless
root@AvejLanjekar:~# docker container ls -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS      NAMES
921393580e18   httpd     "httpd-foreground"      17 minutes ago    Up About a minute    80/tcp     httpd-unless
e0ed2dd929db   httpd     "httpd-foreground"      18 minutes ago    Exited (0) 10 seconds ago              httpd-always
3b1987d5c0e8   httpd     "httpd-foreground"      20 minutes ago    Exited (0) 8 minutes ago              httpd-failure
82b7ef5d2e8f   httpd     "httpd-foreground"      21 minutes ago    Exited (0) 15 minutes ago              httpd-no
root@AvejLanjekar:~#
root@AvejLanjekar:~#
root@AvejLanjekar:~# systemctl restart docker
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS      NAMES
921393580e18   httpd     "httpd-foreground"      17 minutes ago    Up 4 seconds        80/tcp     httpd-unless
e0ed2dd929db   httpd     "httpd-foreground"      19 minutes ago    Up 4 seconds        80/tcp     httpd-always
root@AvejLanjekar:~# docker container ls -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS      NAMES
921393580e18   httpd     "httpd-foreground"      17 minutes ago    Up 7 seconds        80/tcp     httpd-unless
e0ed2dd929db   httpd     "httpd-foreground"      19 minutes ago    Up 7 seconds        80/tcp     httpd-always
3b1987d5c0e8   httpd     "httpd-foreground"      20 minutes ago    Exited (0) 8 minutes ago              httpd-failure
82b7ef5d2e8f   httpd     "httpd-foreground"      21 minutes ago    Exited (0) 16 minutes ago              httpd-no
root@AvejLanjekar:~#
```

Restart Policy 4 :- unless-stopped

```
root@AvejLanjekar:~# docker stop httpd-unless
httpd-unless
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
e0ed2dd929db   httpd     "httpd-foreground"      21 minutes ago    Up 2 minutes    80/tcp    httpd-always
root@AvejLanjekar:~# docker container ls -a
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
921393580e18   httpd     "httpd-foreground"      20 minutes ago    Exited (0) 20 seconds ago    80/tcp    httpd-unless
e0ed2dd929db   httpd     "httpd-foreground"      21 minutes ago    Up 2 minutes    80/tcp    httpd-always
3b1987d5c0e8   httpd     "httpd-foreground"      23 minutes ago    Exited (0) 10 minutes ago    httpd-failure
82b7ef5d2e8f   httpd     "httpd-foreground"      24 minutes ago    Exited (0) 18 minutes ago    httpd-no
root@AvejLanjekar:~# |

root@AvejLanjekar:~#
root@AvejLanjekar:~# systemctl restart docker
root@AvejLanjekar:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
e0ed2dd929db   httpd     "httpd-foreground"      22 minutes ago    Up 4 seconds    80/tcp    httpd-always
root@AvejLanjekar:~# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
921393580e18   httpd     "httpd-foreground"      21 minutes ago    Exited (0) About a minute ago    80/tcp    httpd-unless
e0ed2dd929db   httpd     "httpd-foreground"      22 minutes ago    Up 7 seconds    80/tcp    httpd-always
3b1987d5c0e8   httpd     "httpd-foreground"      23 minutes ago    Exited (0) 11 minutes ago    httpd-failure
82b7ef5d2e8f   httpd     "httpd-foreground"      25 minutes ago    Exited (0) 19 minutes ago    httpd-no
root@AvejLanjekar:~# |
```

Since the container was manually stopped, Docker remembers this and does not restart it after the daemon restarts.

19. Launch an NGINX container with a meaningful name and expose it on the host's port 80. Create an "index.html" file containing the text "Hello there, Let's be the Team CloudEthiX," and copy the file to the container's "/usr/share/nginx/html/" location. Access the container in a browser to verify that the webpage displays correctly.

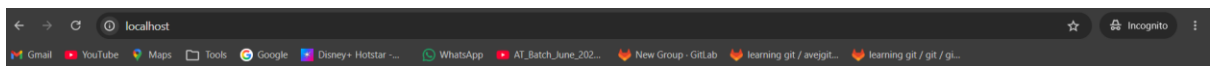
Step 1:- Launch an NGINX container with a meaningful name and expose it on the host's port 80. Create an "index.html" file containing the text "Hello there, Let's be the Team CloudEthiX,"

```
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker run -d --name web-server -p 80:80 nginx
c8f8dc8b4ff80abd02dab826995a3fa207fefff87e8fc186631cc7bfbf3472fcc
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
c8f8dc8b4ff80   nginx     "/docker-entrypoint..."  5 seconds ago    Up 4 seconds    0.0.0.0:80->80/tcp, [::]:80->80/tcp    web-server
e0ed2dd929db   httpd     "httpd-foreground"      30 minutes ago    Up 8 minutes    80/tcp    httpd-always
root@AvejLanjekar:~#
root@AvejLanjekar:~#
root@AvejLanjekar:~# vim index.html
root@AvejLanjekar:~# cat index.html
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX</title>
</head>
<body>
  <h1>Hello there, Let's be the Team CloudEthiX</h1>
</body>
</html>
root@AvejLanjekar:~#
```

Step 2:- copy the file to the container's "/usr/share/nginx/html/" location.

```
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker cp index.html web-server:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to web-server:/usr/share/nginx/html/index.html
root@AvejLanjekar:~# docker exec -it web-server /bin/bash
root@c8fdc8b4ff80:/# cd /usr/share/nginx/html
root@c8fdc8b4ff80:/usr/share/nginx/html# ls
50x.html index.html
root@c8fdc8b4ff80:/usr/share/nginx/html# cat index.html
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX</title>
</head>
<body>
  <h1>Hello there, Let's be the Team CloudEthiX</h1>
</body>
</html>
root@c8fdc8b4ff80:/usr/share/nginx/html#
root@c8fdc8b4ff80:/usr/share/nginx/html#
root@c8fdc8b4ff80:/usr/share/nginx/html#
root@c8fdc8b4ff80:/usr/share/nginx/html# |
```

Step 3:- Access the container in a browser to verify that the webpage displays correctly.



Hello there, Let's be the Team CloudEthiX

20. Pull the Redis image and tag it as "yourname_redis:version 1." Create two additional tags, "yourname_redis:version 2" and "yourname_redis:version 3." Run two Docker containers from the tagged images named "redis01" and "redis02," then delete the "yourname_redis:version 2" image.

Step1 :- Pull the Redis image

```
root@AvejLanjekar:~# docker pull redis
Using default tag: latest
latest: Pulling from library/redis
8f9da2014a53: Pull complete
808f9d34b05d: Pull complete
7a3a8d0170a7: Pull complete
4f4fb700ef54: Pull complete
87fdd3403791: Pull complete
239f6712f038: Pull complete
f37527dca501: Download complete
32e04edd155a: Download complete
Digest: sha256:a7efd4ed5037dcfe1543c0e50b9f06293d3e88827005544bf37f381eea5f01cf
Status: Downloaded newer image for redis:latest
docker.io/library/redis:latest
root@AvejLanjekar:~# docker images
```

IMAGE	ID	DISK	USAGE	CONTENT SIZE	EXTRA
avejlanjekar45/nginx-img:latest	5c96dbbd3eef		238MB	63.1MB	
avejlanjekar45/nginx-img:tagname	5c96dbbd3eef		238MB	63.1MB	
avejlanjekar45/ubuntu-nginx:latest	aeb87ba1ca5e		334MB	85MB	
avejlanjekar45/ubuntu-nginx:v1	aeb87ba1ca5e		334MB	85MB	
httpd:latest	305fd8326a27		177MB	47.6MB	U
lancachenet/ubuntu:latest	f5fdc792dd35		267MB	66MB	
nginx-img:latest	5c96dbbd3eef		238MB	63.1MB	
nginx:latest	5a88c9c45479		241MB	66MB	U
redis:latest	a7efd4ed5037		212MB	57.4MB	
sample-img:latest	fbeb49a8f9fc		238MB	63.1MB	
ubuntu-nginx:latest	aeb87ba1ca5e		334MB	85MB	
ubuntu:latest	3131b4cc82a7		161MB	45.3MB	

```
root@AvejLanjekar:~#
```

Step2:- tag it as "yourname_redis:version 1." Create two additional tags, "yourname_redis:version 2" and "yourname_redis:version 3."

```
root@AvejLanjekar:~# docker tag redis:latest avejlanjekar_redis:version1
root@AvejLanjekar:~# docker tag redis:latest avejlanjekar_redis:version2
root@AvejLanjekar:~# docker tag redis:latest avejlanjekar_redis:version3
root@AvejLanjekar:~#
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker image la
docker: unknown command: docker image la

Usage:  docker image

Run 'docker image --help' for more information
root@AvejLanjekar:~# docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
avejlanjekar45/nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/nginx-img:tagname	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
avejlanjekar45/ubuntu-nginx:v1	aeb87ba1ca5e	334MB	85MB	
avejlanjekar_redis:version1	a7efd4ed5037	212MB	57.4MB	
avejlanjekar_redis:version2	a7efd4ed5037	212MB	57.4MB	
avejlanjekar_redis:version3	a7efd4ed5037	212MB	57.4MB	
httpd:latest	305fd8326a27	177MB	47.6MB	U
lancachenet/ubuntu:latest	f5fdc792dd35	267MB	66MB	
nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
nginx:latest	5a88c9c45479	241MB	66MB	U
redis:latest	a7efd4ed5037	212MB	57.4MB	
sample-img:latest	fbeb49a8f9fc	238MB	63.1MB	
ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
ubuntu:latest	3131b4cc82a7	161MB	45.3MB	

```
root@AvejLanjekar:~# |
```

Step 3:- Run two Docker containers from the tagged images named "redis01" and "redis02"

```
root@AvejLanjekar:~# docker container run -d --name redis01 avejlanjekar_redis:version1
1b142c80d30b81ab70c93269c056db090409079e45bf14d9d3a06fbdcfded5e5
root@AvejLanjekar:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
1b142c80d30b	avejlanjekar_redis:version1	"docker-entrypoint.s..."	4 seconds ago	Up 3 seconds	6379/tcp	redis01
e0ed2dd929db	httpd	"httpd-foreground"	14 hours ago	Up 10 minutes	80/tcp	httpd-a

```
always
root@AvejLanjekar:~# docker container run -d --name redis02 avejlanjekar_redis:version3
5b609339111106f4591335825d99eb90d07cb9c4fff16d5035251e50ab921f1d
root@AvejLanjekar:~# docker container ls
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
5b6093391111	avejlanjekar_redis:version3	"docker-entrypoint.s..."	8 seconds ago	Up 8 seconds	6379/tcp	redis0
2						
1b142c80d30b	avejlanjekar_redis:version1	"docker-entrypoint.s..."	48 seconds ago	Up 48 seconds	6379/tcp	redis0
1						
e0ed2dd929db	httpd	"httpd-foreground"	14 hours ago	Up 10 minutes	80/tcp	httpd-

```
always
root@AvejLanjekar:~#
```

Step 4:- delete the "yourname_redis:version 2" image.

```
root@AvejLanjekar:~# docker image rmi avejlanjekar_redis:version2
Untagged: avejlanjekar_redis:version2
root@AvejLanjekar:~# docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
avejlanjekar45/nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/nginx-img:tagname	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
avejlanjekar45/ubuntu-nginx:v1	aeb87ba1ca5e	334MB	85MB	
avejlanjekar_redis:version1	a7efd4ed5037	212MB	57.4MB	U
avejlanjekar_redis:version3	a7efd4ed5037	212MB	57.4MB	U
httpd:latest	305fd8326a27	177MB	47.6MB	U
lancachenet/ubuntu:latest	f5fdc792dd35	267MB	66MB	
nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
nginx:latest	5a88c9c45479	241MB	66MB	U
redis:latest	a7efd4ed5037	212MB	57.4MB	U
sample-img:latest	fbeb49a8f9fc	238MB	63.1MB	
ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
ubuntu:latest	3131b4cc82a7	161MB	45.3MB	

```
root@AvejLanjekar:~# |
```

- 21. Clone the GitHub - docker-library/httpd: Docker Official Image packaging for Apache HTTP Server Git repository locally and build the httpd Docker image from the Dockerfile at the "httpd/2.4/Dockerfile" path. Run and expose a Docker container from this image with a meaningful name on port 8181, then access the webpage in a browser. Afterward, tag the image as V1.**

Step 1:- Clone the GitHub - docker-library/httpd: Docker Official Image packaging for Apache HTTP Server Git repository locally

```
root@AvejLanjekar:~# git clone https://github.com/docker-library/httpd.git
Cloning into 'httpd'...
remote: Enumerating objects: 1119, done.
remote: Counting objects: 100% (205/205), done.
remote: Compressing objects: 100% (94/94), done.
remote: Total 1119 (delta 144), reused 117 (delta 111), pack-reused 914 (from 4)
Receiving objects: 100% (1119/1119), 227.61 KiB | 1.21 MiB/s, done.
Resolving deltas: 100% (487/487), done.
root@AvejLanjekar:~# ls
compose-demo  docker-compose  docker-project  file.txt  httpd  index.html  test1  ubuntu-nginx
root@AvejLanjekar:~# cd httpd
root@AvejLanjekar:~/httpd# ls
2.4  LICENSE  README.md  generate-stackbrew-library.sh  update.sh
root@AvejLanjekar:~/httpd# cd 2.4
root@AvejLanjekar:~/httpd/2.4# ls
Dockerfile  alpine  httpd-foreground
```

Step2 :- build the httpd Docker image from the Dockerfile at the "httpd/2.4/Dockerfile" path

```
root@AvejLanjekar:~/httpd/2.4# docker build -t http-server .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

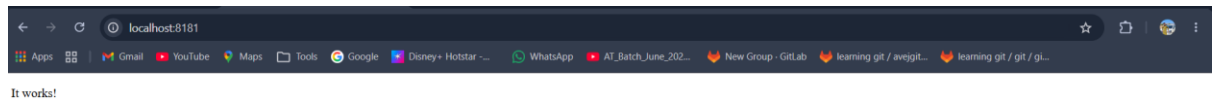
Sending build context to Docker daemon  21.5kB
Step 1/14 : FROM debian:trixie-slim
trixie-slim: Pulling from library/debian
8ff677ca96a9: Download complete
Digest: sha256:020c0d20b9880058cbe785a9db107156c3c75c2ac944a6aa7ab59f2add76a7bd
Status: Downloaded newer image for debian:trixie-slim
--> 020c0d20b988
Step 2/14 : ENV HTTPD_PREFIX /usr/local/apache2
--> Running in 3b07470d7b60
--> Removed intermediate container 3b07470d7b60
--> c5381549d842
Step 3/14 : ENV PATH $HTTPD_PREFIX/bin:$PATH
--> Running in 7d89952d625c
--> Removed intermediate container 7d89952d625c
--> c1537ce331c3
Step 4/14 : RUN mkdir -p "$HTTPD_PREFIX"      && chown www-data:www-data "$HTTPD_PREFIX"
--> Running in 9abb0bac20f0
--> Removed intermediate container 9abb0bac20f0
--> 2dfc8b26320e
Step 5/14 : WORKDIR $HTTPD_PREFIX
--> Running in b26e8f5f9dbb
--> Removed intermediate container b26e8f5f9dbb
--> f35fdeefd716
Step 6/14 : RUN set -eux; apt-get install --update -y --no-install-recommends ca-certificates
ibaprutil1-ldap libldap-common ; apt-get dist-clean
```

Step 3:- Run and expose a Docker container from this image with a meaningful name on port 8181

```
root@AvejLanjekar:~/httpd/2.4#
root@AvejLanjekar:~/httpd/2.4# docker container run -d --name webapp -p 8181:80 httpd-server
Unable to find image 'httpd-server:latest' locally
docker: Error response from daemon: pull access denied for httpd-server, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
root@AvejLanjekar:~/httpd/2.4# docker container run -d --name webapp -p 8181:80 http-server:latest
998d40cba39bf60513976397bb3aa2e83e88e1b474d018f91f095e2de3193014
root@AvejLanjekar:~/httpd/2.4#
root@AvejLanjekar:~/httpd/2.4#
root@AvejLanjekar:~/httpd/2.4#
root@AvejLanjekar:~/httpd/2.4# docker container ls
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
998d40cba39b   http-server:latest                 "httpd-foreground"      6 seconds ago Up 5 seconds  0.0.0.0:8181->80/
5b6093391111   avejlanjekar_redis:version3       "docker-entrypoint.s..." 19 minutes ago Up 19 minutes  6379/tcp
1b142c80d30b   avejlanjekar_redis:version1       "docker-entrypoint.s..." 19 minutes ago Up 19 minutes  6379/tcp
e0ed2dd929db   httpd                              "httpd-foreground"      14 hours ago   Up 29 minutes  80/tcp
```

Step 5:- access the webpage in a browser



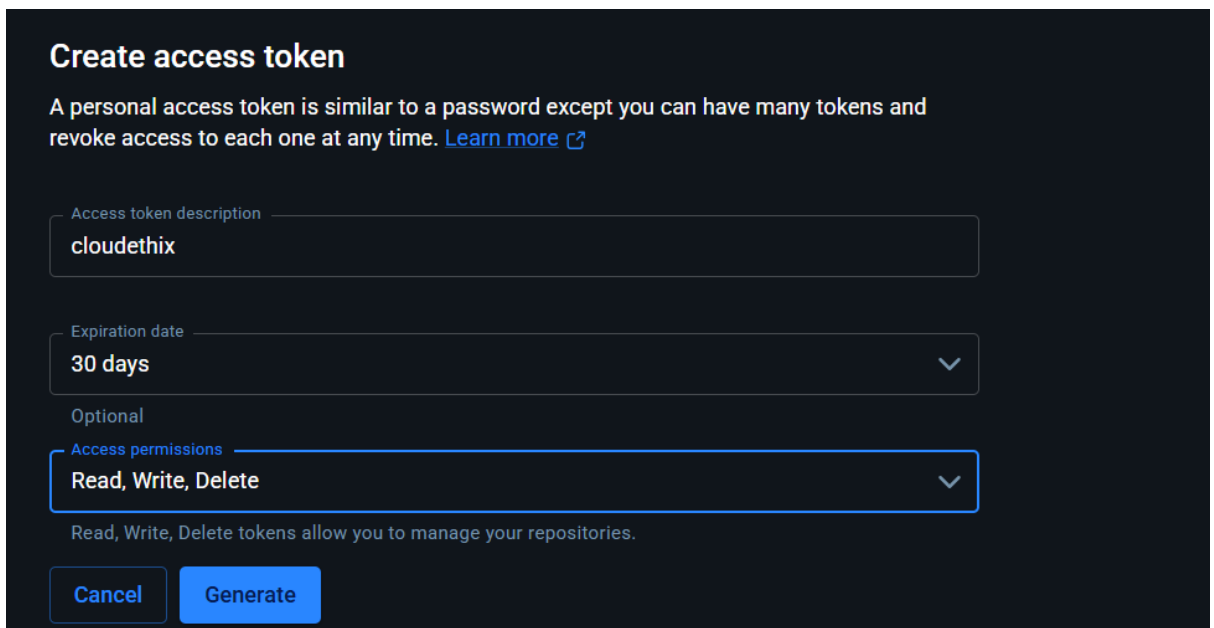
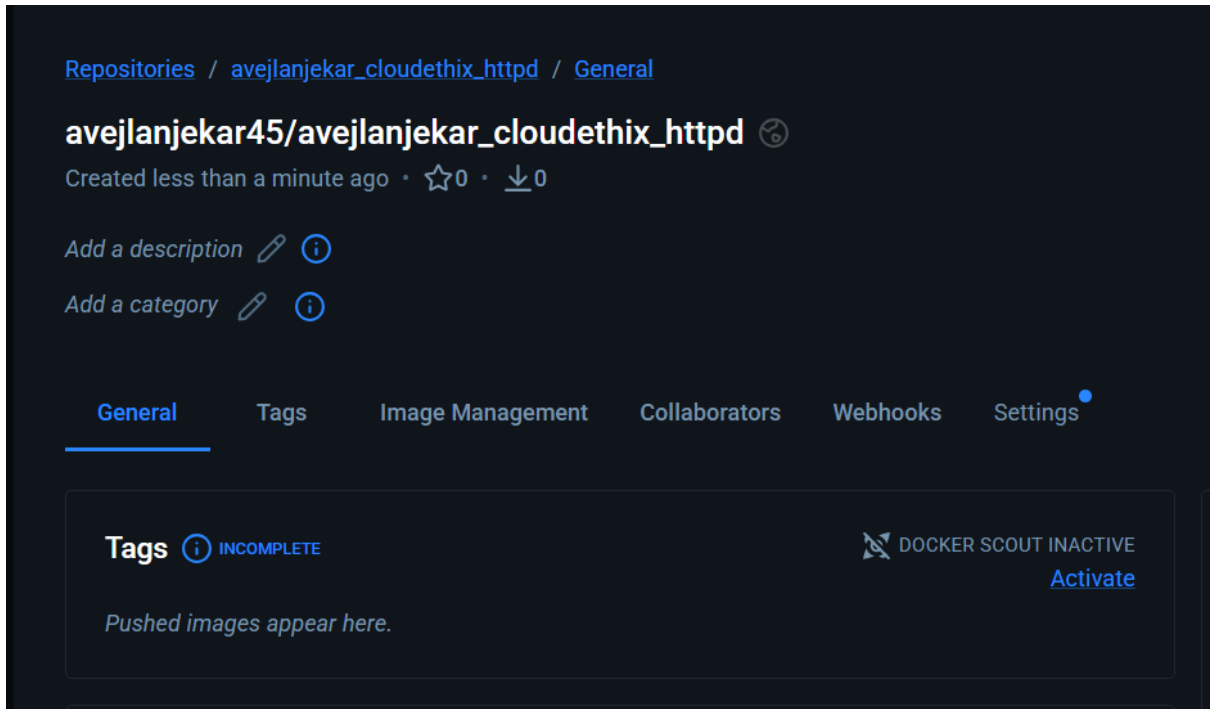
Step 6 :- tag the image as V1

```
root@AvejLanjekar:~/httpd/2.4#
root@AvejLanjekar:~/httpd/2.4# docker tag http-server:latest http-server:v1
root@AvejLanjekar:~/httpd/2.4# docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
avejlanjekar45/nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/nginx-img:tagname	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
avejlanjekar45/ubuntu-nginx:v1	aeb87ba1ca5e	334MB	85MB	
avejlanjekar_redis:version1	a7efd4ed5037	212MB	57.4MB	U
avejlanjekar_redis:version3	a7efd4ed5037	212MB	57.4MB	U
debian:trixie-slim	020c0d20b988	119MB	31.8MB	
http-server:latest	81e6a2a737aa	175MB	45.3MB	U
http-server:v1	81e6a2a737aa	175MB	45.3MB	U
httpd:latest	305fd8326a27	177MB	47.6MB	U
lancachenet/ubuntu:latest	f5fdc792dd35	267MB	66MB	
nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
nginx:latest	5a88c9c45479	241MB	66MB	U
redis:latest	a7efd4ed5037	212MB	57.4MB	U
sample-img:latest	fbeb49a8f9fc	238MB	63.1MB	
ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
ubuntu:latest	3131b4cc82a7	161MB	45.3MB	

22. Create a Docker Hub account and generate a security token. Establish a Docker Hub repository named "yourname_cloudethix_httpd." Log in to Docker Hub and push the previously tagged httpd image to your repository.

Step 1:- Create a Docker Hub account and generate a security token . Establish a Docker Hub repository named "yourname_cloudethix_httpd."



Step 2:- Log in to Docker Hub

```
root@AvejLanjekar:~/httpd/2.4#  
root@AvejLanjekar:~/httpd/2.4# docker login -u avejlanjekar45  
  
Info + A Personal Access Token (PAT) can be used instead.  
To create a PAT, visit https://app.docker.com/settings  
  
Password:  
  
WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.  
Configure a credential helper to remove this warning. See  
https://docs.docker.com/go/credential-store/  
  
Login Succeeded  
root@AvejLanjekar:~/httpd/2.4#
```


Step 3:- push the previously tagged httpd image to your repository.

```
root@Avejlanjekar:~/httpd/2.4# docker tag http-server:v1 avejlanjekar45/avejlanjekar_cloudethix_httpd:v1
root@Avejlanjekar:~/httpd/2.4# docker push avejlanjekar45/avejlanjekar_cloudethix_httpd:v1
The push refers to repository [docker.io/avejlanjekar45/avejlanjekar_cloudethix_httpd]
363414dc189d: Layer already exists
c9af93e60144: Layer already exists
a57c7ac74d0b: Layer already exists
6fd03ddfbac3: Layer already exists
062e450697fa: Layer already exists
v1: digest: sha256:81e6a2a737aabfe96131f737f452b76055777c8ac50fbbeb03e1fd7a4c7c8c84 size: 1403
root@Avejlanjekar:~/httpd/2.4#
```



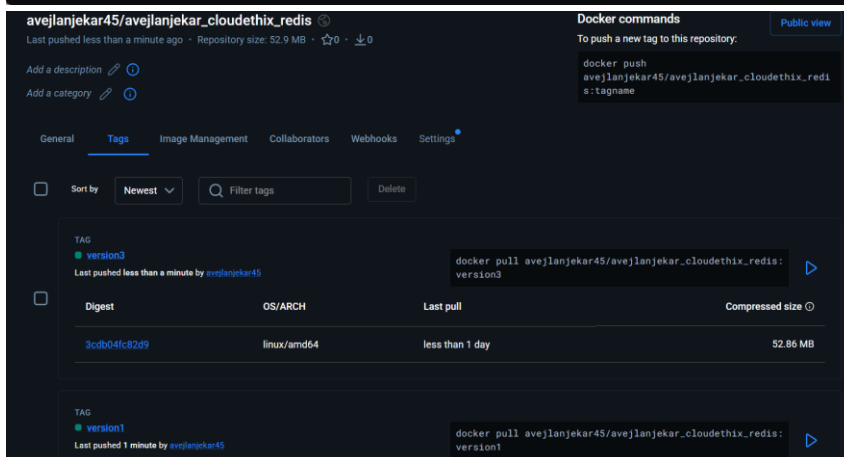
23. Push the Redis images tagged as version 1 and 3 to your Docker Hub repository, named "yourname_cloudethix_redis."

```
root@Avejlanjekar:~/httpd/2.4# docker tag avejlanjekar_redis:version1 avejlanjekar45/avejlanjekar_cloudethix_redis:version1
root@Avejlanjekar:~/httpd/2.4# docker tag avejlanjekar_redis:version3 avejlanjekar45/avejlanjekar_cloudethix_redis:version3
```

```
root@Avejlanjekar:~/httpd/2.4# docker push avejlanjekar45/avejlanjekar_cloudethix_redis:version1
The push refers to repository [docker.io/avejlanjekar45/avejlanjekar_cloudethix_redis]
8f9da2014a53: Mounted from library/redis
239f6712f038: Mounted from library/redis
7a3a8d0170a7: Mounted from library/redis
062e450697fa: Mounted from avejlanjekar45/http-server
87fdd3403791: Mounted from library/redis
808f9d34b05d: Mounted from library/redis
4f4fb708ef5d: Mounted from library/redis
version1: digest: sha256:3cdb04fc82d909d929d4fcf30ee8011491a8dcc0a91a44b8757fa13077d0e06f size: 2287

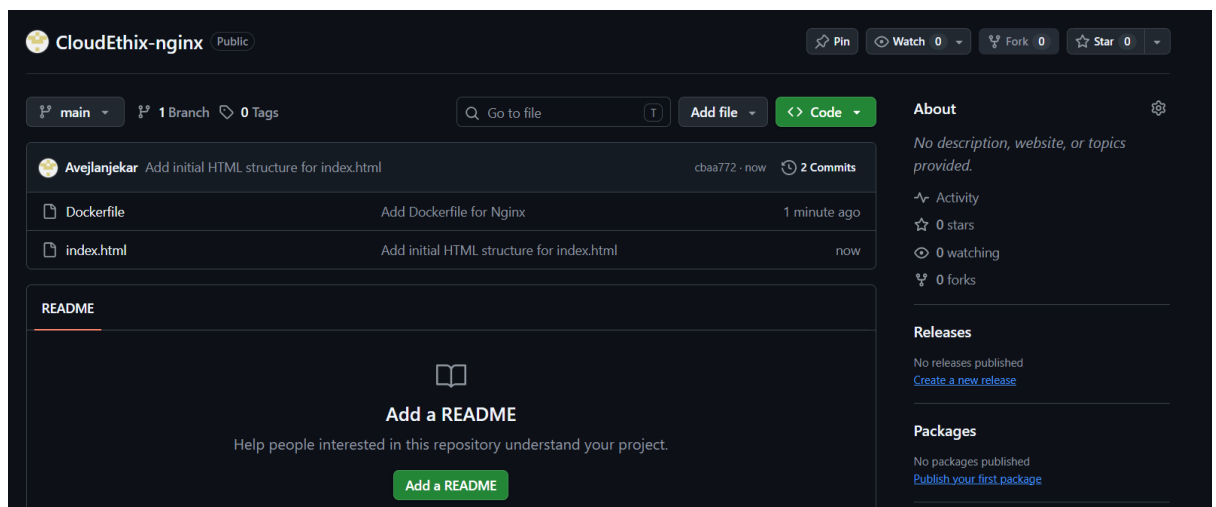
Info → Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:a7ef44ed507dcfe1543c0e50b9f06293d3e8827005544bf37f381eed5f01cf -> sha256:3cdb04fc82d909d929d4fcf30ee8011491a8dcc0a91a44b8757fa13077d0e06f
root@Avejlanjekar:~/httpd/2.4# docker push avejlanjekar45/avejlanjekar_cloudethix_redis:version3
The push refers to repository [docker.io/avejlanjekar45/avejlanjekar_cloudethix_redis]
8f9da2014a53: Layer already exists
239f6712f038: Layer already exists
808f9d34b05d: Layer already exists
4f4fb708ef5d: Layer already exists
062e450697fa: Layer already exists
87fdd3403791: Layer already exists
7a3a8d0170a7: Layer already exists
version3: digest: sha256:3cdb04fc82d909d929d4fcf30ee8011491a8dcc0a91a44b8757fa13077d0e06f size: 2287

Info → Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:a7ef44ed507dcfe1543c0e50b9f06293d3e8827005544bf37f381eed5f01cf -> sha256:3cdb04fc82d909d929d4fcf30ee8011491a8dcc0a91a44b8757fa13077d0e06f
root@Avejlanjekar:~/httpd/2.4#
```



- 24. Create a remote Git repository and add the Dockerfile and index.html files. Clone the repository locally and create a release branch. Build the Docker image from the release branch with a meaningful tag, then run a container from the image and expose it on host port 8383. Check the webpage in a browser, and upon success, push the image to your Docker Hub repository named "yourname_cloudethix_nginx." Finally, push the release branch to the remote Git repository .**

Step 1:- Create a remote Git repository and add the Dockerfile and index.html files



Step 2:- Clone the repository locally

```
root@AvejLanjekar:~/httpd/2.4# git clone https://github.com/Avejlanjekar/CloudEthix-nginx.git
Cloning into 'CloudEthix-nginx'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
root@AvejLanjekar:~/httpd/2.4# ls
CloudEthix-nginx  Dockerfile  alpine  httpd-foreground
root@AvejLanjekar:~/httpd/2.4# cd CloudEthix-nginx/
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# ls
Dockerfile  index.html
```

Step 3:- create a release branch. Build the Docker image from the release branch with a meaningful tag, then run a container from the image and expose it on host port 8383

```
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# git branch
* main
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# git branch release
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# git branch
* main
  release
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# git checkout release
Switched to branch 'release'
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# git branch
main
* release
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# ls
Dockerfile  index.html
```

Build the Docker image from the release branch with a meaningful tag

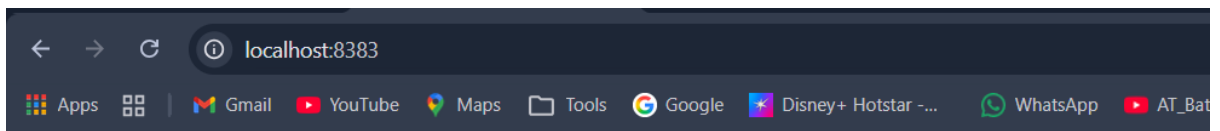
```
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# docker build -t cloudeethix-nginx .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 66.56kB
Step 1/3 : FROM nginx:latest
----> 5a88c9c45479
Step 2/3 : COPY index.html /usr/share/nginx/html/index.html
----> Using cache
----> e68918691ed8
Step 3/3 : EXPOSE 80
----> Using cache
----> 7cb1a3281b3d
Successfully built 7cb1a3281b3d
Successfully tagged cloudeethix-nginx:latest
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx#
```

run a container from the image and expose it on host port 8383

```
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# docker run -d -p 8383:80 --name cloudeethix-nginx-cont cloudeethix-nginx
e559b2b4df24dc4b903af3965a129564c8873bee6c013c8bfff28b16fff659dc7e
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# docker container ls
CONTAINER ID   IMAGE                                COMMAND                  CREATED          STATUS          PORTS
e559b2b4df24   cloudeethix-nginx                  "/docker-entrypoint..." 6 seconds ago    Up 5 seconds    0.0.0.0:8383->80/tcp, [::]:8383->80/tcp
cloudeethix-nginx-cont
```

Step 4:- . Check the webpage in a browser

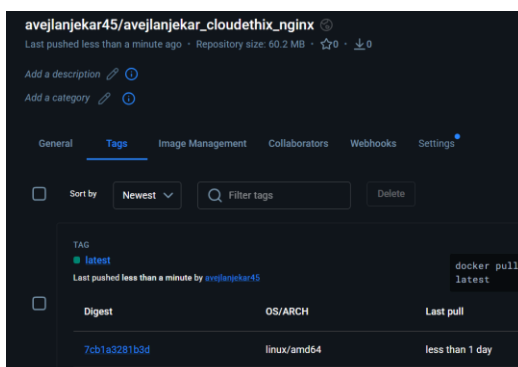


Welcome to CloudEthix Nginx Docker Container

Step 5:- push the image to your Docker Hub repository named

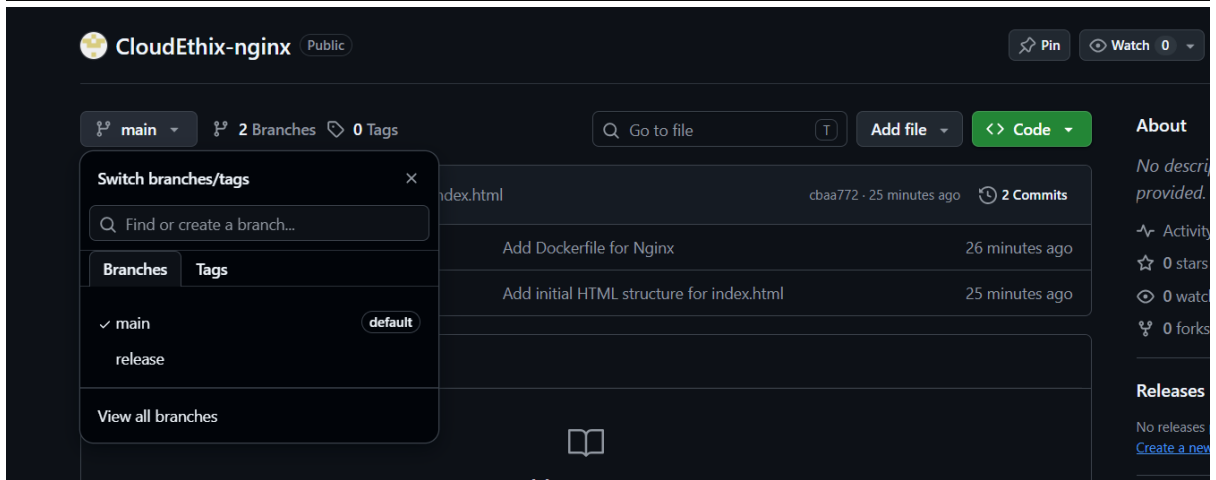
"yourname_cloudeethix_nginx"

```
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# docker tag cloudeethix-nginx:latest avejlanjekar45/avejlanjekar_cloudeethix_nginx
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# docker push avejlanjekar45/avejlanjekar_cloudeethix_nginx
Using default tag: latest
The push refers to repository [docker.io/avejlanjekar45/avejlanjekar_cloudeethix_nginx]
82454cdbf456: Mounted from library/nginx
b6698f04e005: Mounted from library/nginx
b3c8622045bf: Pushed
cacfcdd01f30: Mounted from library/nginx
2bedaf25031a: Mounted from library/nginx
d26f27cc8c41: Mounted from library/nginx
062e450697fa: Mounted from avejlanjekar45/http-server
3c7ab7949321: Mounted from library/nginx
latest: digest: sha256:7cb1a3281b3d43bcbbf59b1545c0d94fd621994cae2ca1ea9b78d09a5f43387c size: 2043
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx#
```



Step 6:- Finally, push the release branch to the remote Git repository .

```
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx#
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx#
root@AvejLanjekar:~/httpd/2.4/CloudEthix-nginx# git push origin release
```



25. Save all local Redis images as a .tar file in the master branch of your local repository. Delete all Redis images from your local system and push the master branch to the remote repository. Load the Redis images from the tar file to your local system, and verify that all images have been loaded correctly.

Step 1 :- Save all local Redis images as a .tar file in the master branch of your local repository

```
root@AvejLanjekar:~/CloudEthix-nginx# docker save -o redis-images.tar avejlanjekar45/avejlanjekar_cloudethix_redis:version1 avejlanjekar45/avejlanjekar_cloudethix_redis:version3 redis:latest
root@AvejLanjekar:~/CloudEthix-nginx# ls
Dockerfile  index.html  redis-images.tar
root@AvejLanjekar:~/CloudEthix-nginx# git add redis-images.tar
root@AvejLanjekar:~/CloudEthix-nginx# git commit -m "Added Added Redis image Backup"
[main 3354e39] Added Added Redis image Backup
  Committer: root <root@AvejLanjekar.localdomain>
  Your name and email address were configured automatically based
  on your username and hostname. Please check that they are accurate.
  You can suppress this message by setting them explicitly. Run the
  following command and follow the instructions in your editor to edit
  your configuration file:
    git config --global --edit
  After doing this, you may fix the identity used for this commit with:
    git commit --amend --reset-author
  1 file changed, 0 insertions(+), 0 deletions(-)
  create mode 100644 redis-images.tar
root@AvejLanjekar:~/CloudEthix-nginx# git status
On branch main
nothing to commit, working tree clean
```

Step 2:- Delete all Redis images from your local system and push the master branch to the remote repository

```
root@AvejLanjekar:~/CloudEthix-nginx# docker rmi -f redis:latest avejlanjekar45/avejlanjekar_cloudethix_redis:version3 avejlanjekar45/avejlanjekar_cloudethix_redis:version1
Untagged: redis:latest
Deleted: sha256:a7efd4ed5837dcfe1543ce50b9f86293d3e88827085544bf37f381eea5f01cf
Untagged: avejlanjekar45/avejlanjekar_cloudethix_redis:version3
Untagged: avejlanjekar45/avejlanjekar_cloudethix_redis:version1
Deleted: sha256:3cdb04fc824909d929d4fc30ee8011491a8dccc8a91a44b8757fa13077d0e06f
```

```

root@AvejLanjekar:~/CloudEthix-nginx# git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 52.75 MiB | 4.10 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: warning: See https://gh.io/lfs for more information.
remote: warning: File redis-images.tar is 54.81 MB; this is larger than GitHub's recommended maximum file size of 50.00 MB
remote: warning: GH001: Large files detected. You may want to try Git Large File Storage - https://git-lfs.github.com.
To github.com:AvejLanjekar/CloudEthix-nginx.git
25363bb..3354c39 main -> main
root@AvejLanjekar:~/CloudEthix-nginx#

```

Step 3:- Load the Redis images from the tar file to your local system, and verify that all images have been loaded correctly.

```

root@AvejLanjekar:~/CloudEthix-nginx# docker load -i redis-images.tar
loaded image: avejlanjekar45/avejlanjekar_cloudethix_redis:version1
loaded image: avejlanjekar45/avejlanjekar_cloudethix_redis:version3
loaded image: redis:latest
root@AvejLanjekar:~/CloudEthix-nginx# docker image ls

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
avejlanjekar45/avejlanjekar_cloudethix_httpd:v1	81e6a2a737aa	175MB	45.3MB	U
avejlanjekar45/avejlanjekar_cloudethix_nginx:latest	7cbl1a3281b3d	238MB	63.1MB	U
avejlanjekar45/avejlanjekar_cloudethix_redis:version1	3cdb04fc82d9	210MB	55.4MB	
avejlanjekar45/avejlanjekar_cloudethix_redis:version3	3cdb04fc82d9	210MB	55.4MB	
avejlanjekar45/nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/nginx-img:tagname	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
avejlanjekar45/ubuntu-nginx:v1	aeb87ba1ca5e	334MB	85MB	
cloudethix-nginx:latest	7cbl1a3281b3d	238MB	63.1MB	U
debian:trixie-slim	020c0d20b988	119MB	31.8MB	
http-server:latest	81e6a2a737aa	175MB	45.3MB	U
http-server:v1	81e6a2a737aa	175MB	45.3MB	U
httpd:latest	305fd8326a27	177MB	47.6MB	U
lancachenet/ubuntu:latest	f5fdc792dd35	267MB	66MB	
nginx-container:latest	7cbl1a3281b3d	238MB	63.1MB	U
nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
nginx:latest	5a88c9c45479	241MB	66MB	U
redis:latest	a7efd4ed5037	212MB	57.4MB	
sample-img:latest	fbab49a8f9fc	238MB	63.1MB	
ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
ubuntu:latest	3131b4cc82a7	161MB	45.3MB	

```

root@AvejLanjekar:~/CloudEthix-nginx# |

```

26. Pull the Busy-box image to your local system, tag it, and push it to the Docker Hub repository "yourname_cloudethix_busybox." Export the Docker image from the NGINX container, create a .tar file, and import the tar file to create a Docker image with a meaningful name. After importing the image, tag it and push it to the "yourname_cloudethix_busybox" Docker Hub repository.

Step 1:- Pull the Busy-box image to your local system

```

root@AvejLanjekar:~# docker pull busybox
Using default tag: latest
latest: Pulling from library/busybox
b05093807bb0: Pull complete
7270b3e1860c: Download complete
Digest: sha256:dc2d74b28e4cf8984fa52af1f39bc7c3d9c73760b41a74d629f5d11b1ab28616
Status: Downloaded newer image for busybox:latest
root@AvejLanjekar:~# docker image ls

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
avejlanjekar45/avejlanjekar_cloudethix_httpd:v1	81e6a2a737aa	175MB	45.3MB	U
avejlanjekar45/avejlanjekar_cloudethix_nginx:latest	7cbl1a3281b3d	238MB	63.1MB	U
avejlanjekar45/avejlanjekar_cloudethix_redis:version1	3cdb04fc82d9	210MB	55.4MB	
avejlanjekar45/avejlanjekar_cloudethix_redis:version3	3cdb04fc82d9	210MB	55.4MB	
avejlanjekar45/nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/nginx-img:tagname	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
avejlanjekar45/ubuntu-nginx:v1	aeb87ba1ca5e	334MB	85MB	
busybox:latest	dc2d74b28e4c	6.81MB	2.23MB	
cloudethix-nginx:latest	7cbl1a3281b3d	238MB	63.1MB	U
debian:trixie-slim	020c0d20b988	119MB	31.8MB	
http-server:latest	81e6a2a737aa	175MB	45.3MB	U
http-server:v1	81e6a2a737aa	175MB	45.3MB	U
httpd:latest	305fd8326a27	177MB	47.6MB	U
lancachenet/ubuntu:latest	f5fdc792dd35	267MB	66MB	
nginx-container:latest	7cbl1a3281b3d	238MB	63.1MB	U
nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
nginx:latest	5a88c9c45479	241MB	66MB	U
redis:latest	a7efd4ed5037	212MB	57.4MB	
sample-img:latest	fbab49a8f9fc	238MB	63.1MB	
ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
ubuntu:latest	3131b4cc82a7	161MB	45.3MB	

Step 2:- tag it, and push it to the Docker Hub repository

"yourname_cloudethix_busybox

```
root@AvejLanjekar:~# docker tag busybox:latest avejlanjekar45/avejlanjekar_cloudethix_busybox
root@AvejLanjekar:~# docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
avejlanjekar45/avejlanjekar_cloudethix_busybox:latest	dc2d74b28e4c	6.81MB	2.23MB	

```
root@AvejLanjekar:~# docker push avejlanjekar45/avejlanjekar_cloudethix_busybox
Using default tag: latest
The push refers to repository [docker.io/avejlanjekar45/avejlanjekar_cloudethix_busybox]
b05093807bb0: Mounted from library/busybox
latest: digest: sha256:1cfa4e2b09e127b9c4ed43578d3f3c18e7d44ea47b9ea98475c0cbe9086525f8 size: 610

Info → Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:dc2d74b28e4cf8984fa52af1f39bc7c3d9c73760b41a74d629f5d11b1ab28616 -> sha256:1cfa4e2b09e127b9c4ed43578d3f3c18e7d44ea47b9ea98475c0cbe9086525f8
root@AvejLanjekar:~# |
```

avejlanjekar45/avejlanjekar_cloudethix_busybox

Last pushed less than a minute ago · Repository size: 2.1 MB · 0 · 0

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Tags DOCKER SCOUT INACTIVE [Activate](#)

This repository contains 0 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	less than a minute

[See all](#)

Step 3:- Export the Docker image from the NGINX container, create a .tar file, and import the tar file to create a Docker image with a meaningful name

```
root@AvejLanjekar:~# docker export -o nginx-container.tar cloudethix-nginx-cont
root@AvejLanjekar:~# ls
CloudEthix-nginx  compose-demo  docker-compose  docker-project  file.txt  httpd  index.html  nginx-container.tar  test1  ubuntu-nginx
```

import the tar file to create a Docker image with a meaningful name

```
root@AvejLanjekar:~# docker import nginx-container.tar nginx-imported:v1
sha256:c7133eedb282e956b5f85dcd2542db247d4baaf0466b415035eef0173494f5f
root@AvejLanjekar:~# docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
avejlanjekar45/avejlanjekar_cloudethix_busybox:latest	dc2d74b28e4c	6.81MB	2.23MB	
avejlanjekar45/avejlanjekar_cloudethix_httpd:v1	81e6a2a737aa	175MB	45.3MB	
avejlanjekar45/avejlanjekar_cloudethix_nginx:latest	7cb1a3281b3d	238MB	63.1MB	
avejlanjekar45/avejlanjekar_cloudethix_redis:version1	3cdb04fc82d9	210MB	55.4MB	
avejlanjekar45/avejlanjekar_cloudethix_redis:version3	3cdb04fc82d9	210MB	55.4MB	
avejlanjekar45/nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/nginx-img:tagname	5c96dbbd3eef	238MB	63.1MB	
avejlanjekar45/ubuntu-nginx:latest	aeb87balca5e	334MB	85MB	
avejlanjekar45/ubuntu-nginx:v1	aeb87balca5e	334MB	85MB	
busybox:latest	dc2d74b28e4c	6.81MB	2.23MB	
cloudethix-nginx:latest	7cb1a3281b3d	238MB	63.1MB	
debian:trixie-slim	020c0d20b988	119MB	31.8MB	
http-server:latest	81e6a2a737aa	175MB	45.3MB	
http-server:v1	81e6a2a737aa	175MB	45.3MB	
httpd:latest	305fd0926a27	177MB	47.6MB	
lancachenet/ubuntu:latest	f5fdc792dd35	267MB	66MB	
nginx-container:latest	7cb1a3281b3d	238MB	63.1MB	
nginx-img:latest	5c96dbbd3eef	238MB	63.1MB	
nginx-imported:v1	c7133eedb282	236MB	63.7MB	
nginx:latest	5a88c9c45479	241MB	66MB	
redis:latest	a7efd4ed5037	212MB	57.4MB	
sample-img:latest	fbab49a8f9fc	238MB	63.1MB	
ubuntu-nginx:latest	aeb87balca5e	334MB	85MB	
ubuntu:latest	3131b4cc82a7	161MB	45.3MB	

Step 4:- push it to the "yourname_cloudethix_busybox" Docker Hub repository.

```
root@AvejLanjekar:~# docker tag nginx-imported:v1 avejlanjekar45/avejlanjekar_cloudethix_busybox:imported
root@AvejLanjekar:~# docker push avejlanjekar45/avejlanjekar_cloudethix_busybox:imported
The push refers to repository [docker.io/avejlanjekar45/avejlanjekar_cloudethix_busybox]
25fb47e3db48: Mounted from avejlanjekar45/avejlanjekar45_cloudethix_busybox
imported: digest: sha256:c7133eedb282e956b5f85dcd2542db247d4baaf0466b415035eef0173494f5f size: 406
root@AvejLanjekar:~# |
```


27. Dockerfile creation: Create a simple Dockerfile to build a custom image with the following specifications:

Base image: Ubuntu

Install packages: curl, vim, and git

Set an environment variable: MY_NAME=Your_Name

Build the custom image using docker build and list all available images using docker images

```
root@AvejLanjekar:~# mkdir Built-Dockerfile
root@AvejLanjekar:~# cd Built-Dockerfile/
root@AvejLanjekar:~/Built-Dockerfile# ls
root@AvejLanjekar:~/Built-Dockerfile# vim Dockerfile
root@AvejLanjekar:~/Built-Dockerfile#
```

```
root@AvejLanjekar:~/Built-Dockerfile# cat Dockerfile

FROM ubuntu

RUN apt-get update && \
    apt-get install -y curl vim git && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

ENV My_name="Avej lanjekar"

CMD ["/bin/bash"]

root@AvejLanjekar:~/Built-Dockerfile# |
```

```
root@AvejLanjekar:~/Built-Dockerfile# docker build -t ubuntu-custom:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.048KB
Step 1/4 : FROM ubuntu
--> 3131b4cc82a7
Step 2/4 : RUN apt-get update && apt-get install -y curl vim git && apt-get clean && rm -rf /var/lib/apt/lists/*
--> Running in 05ec5a9f73bf
```

```
root@AvejLanjekar:~/Built-Dockerfile# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
avejlanjekar45/avejlanjekar_cloudethix_busybox:imported	c7133e6db282	236MB	63.7MB	
avejlanjekar45/avejlanjekar_cloudethix_busybox:latest	dc2d74b28e4c	6.81MB	2.23MB	
avejlanjekar45/avejlanjekar_cloudethix_httpd:v1	81e6a2a737aa	175MB	45.3MB	U
avejlanjekar45/avejlanjekar_cloudethix_nginx:latest	7cb1a3281b3d	238MB	63.1MB	
avejlanjekar45/avejlanjekar_cloudethix_redis:version1	3c0b04fc82d9	210MB	55.4MB	
avejlanjekar45/avejlanjekar_cloudethix_redis:version3	3c0b04fc82d9	210MB	55.4MB	
avejlanjekar45/nginx-ing:latest	5c96dbdb3eef	238MB	63.1MB	
avejlanjekar45/nginx-ing:tagname	5c96dbdb3eef	238MB	63.1MB	
avejlanjekar45/ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
avejlanjekar45/ubuntu-nginx:v1	aeb87ba1ca5e	334MB	85MB	
busybox:latest	dc2d74b28e4c	6.81MB	2.23MB	
cloudethix-nginx:latest	7cb1a3281b3d	238MB	63.1MB	U
debian:trixie-slim	020c0d20b988	119MB	31.8MB	
http-server:latest	81e6a2a737aa	175MB	45.3MB	U
http-server:v1	81e6a2a737aa	175MB	45.3MB	U
httpd:latest	305f40832a27	177MB	47.6MB	
lancachenet/ubuntu:latest	f5f9c792dd35	267MB	66MB	
nginx-container:latest	7cb1a3281b3d	238MB	63.1MB	U
nginx-ing:latest	5c96dbdb3eef	238MB	63.1MB	
nginx-imported:v1	c7133e6db282	236MB	63.7MB	
nginx:latest	5a88c9c45479	241MB	66MB	U
redis:latest	a7efd4ed5037	212MB	57.4MB	
sample-ing:latest	f0eb49a859fc	238MB	63.1MB	
ubuntu-custom:v1	4017bd644494	407MB	98.9MB	U
ubuntu-nginx:latest	aeb87ba1ca5e	334MB	85MB	
ubuntu:latest	3131b4cc82a7	161MB	45.3MB	

```
root@AvejLanjekar:~/Built-Dockerfile#
```

- 28. Docker networking: Create two Docker containers with different names, and add them to the same custom network. Verify that the containers can communicate with each other using their names as hostnames.**

Step 1 :- create docker network

```
root@AvejLanjekar:~# docker network create --driver bridge docker-lab
ab3dc90848a6de9f4090e577fb28ca1edfc7a05af60411168005e577d852114d
root@AvejLanjekar:~# docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
6cc8c4715896        alpine-net          bridge              local
d8db692f7366        bridge              bridge              local
18f0646935f9        docker-compose_default bridge              local
ab3dc90848a6        docker-lab          bridge              local
fd609fea3662        host                host                local
732a8745d38e        none                null                local
```

Step 2:- Create two Docker containers with different names, and add them to the same custom network

```
root@AvejLanjekar:~# docker run -d --name nginx-cont1 --network docker-lab nginx
fa248d95f8f0b0f0f04ac757db5a49028d6aa3db7ab3985d0d4a31121c1c8b7
root@AvejLanjekar:~# docker run -d --name nginx-cont2 --network docker-lab nginx
62068bbc5209cb3b97045c1194f7dba49bfff99fb4a1a25f3593ae22dfd2e14
root@AvejLanjekar:~# docker network ls
root@AvejLanjekar:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                    NAMES
62068bbc5209  nginx    "/docker-entrypoint..." 32 seconds ago Up 31 seconds   80/tcp                  nginx-cont2
fa248d95f8f0  nginx    "/docker-entrypoint..." About a minute ago Up About a minute   80/tcp                  nginx-cont1
e559b2b4df24  cloude... "httpd-foreground"      2 hours ago    Up 2 hours      0.0.0.0:8383->80/tcp, [...]8383->80/tcp cloude...nginx-cont
998d40cba39b  http-server:latest "httpd-foreground"      3 hours ago    Up 3 hours      0.0.0.0:8181->80/tcp, [...]8181->80/tcp webapp
e0ed2dd929db  httpd    "httpd-foreground"      16 hours ago   Up 3 hours      80/tcp                  httpd-always
root@AvejLanjekar:~#
```

Step 3:- Verify that the containers can communicate with each other using their names as hostnames.

```
root@AvejLanjekar:~# docker exec -it nginx-cont1 /bin/bash
root@fa248d95f8f0:/# ping nginx-cont2
PING nginx-cont2 (172.19.0.3) 56(84) bytes of data.
64 bytes from nginx-cont2.docker-lab (172.19.0.3): icmp_seq=1 ttl=64 time=1.11 ms
64 bytes from nginx-cont2.docker-lab (172.19.0.3): icmp_seq=2 ttl=64 time=0.802 ms
64 bytes from nginx-cont2.docker-lab (172.19.0.3): icmp_seq=3 ttl=64 time=0.087 ms
64 bytes from nginx-cont2.docker-lab (172.19.0.3): icmp_seq=4 ttl=64 time=0.114 ms
^C
--- nginx-cont2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3030ms
rtt min/avg/max/mdev = 0.087/0.527/1.106/0.440 ms
root@fa248d95f8f0:/#
```

- 29. Data persistence with volumes: Run a Docker container with an attached volume, create a file inside the volume, and verify that the file persists after the container is stopped and removed.**

Step 1:- create volume

```
root@AvejLanjekar:~# docker volume create cloude...-volume
cloude...-volume
root@AvejLanjekar:~# docker volume ls
DRIVER      VOLUME NAME
local      cloude...-volume
local      data-vol
local      data-volume
local      docker-compose_nginx-data
local      my-volume
local      nginx-volume
local      shared-data
root@AvejLanjekar:~#
```



```
root@AvejLanJekar:~# docker run -d --name mynginx -v cloudethix-volume:/usr/share/nginx/html -p 8089:80 nginx
3572a88a49948580f51fb33ceaef05bade8bffa47b8504b2c0976295e30a5
root@AvejLanJekar:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
3572a88a4994	nginx	"docker-entrypoint..."	3 seconds ago	Up 2 seconds	0.0.0.0:8089->80/tcp, [::]:8089->80/tcp	mynginx
62068bb5c209	nginx	"docker-entrypoint..."	17 minutes ago	Up 17 minutes	80/tcp	nginx-cont2
fa248d95f8f0	nginx	"docker-entrypoint..."	17 minutes ago	Up 17 minutes	80/tcp	nginx-cont1
e559b2b4df24	cloudethix-nginx	"docker-entrypoint..."	2 hours ago	Up 2 hours	0.0.0.0:8383->80/tcp, [::]:8383->80/tcp	cloudethix-nginx-cont
998d4c3a39b	http-server:latest	"httpd-foreground"	3 hours ago	Up 3 hours	0.0.0.0:8181->80/tcp, [::]:8181->80/tcp	webapp
e0e2dd2929b	httpd	"httpd-foreground"	17 hours ago	Up 3 hours	80/tcp	httpd-always

```
root@AvejLanJekar:~#
root@AvejLanJekar:~#
root@AvejLanJekar:~# docker exec -it mynginx /bin/bash
root@3572a88a4994:/# echo "hello cloudethix , be always team cloudethix" > /usr/share/nginx/html/index.html
root@3572a88a4994:/# exit
exit
root@AvejLanJekar:~# docker stop mynginx
mynginx
root@AvejLanJekar:~# docker rm mynginx
mynginx
root@AvejLanJekar:~#
```

```

root@AveJlanjekar:~# docker run -d --name cloudeithx-mynginx -p 8090:80 -v cloudeithx-volume:/usr/share/nginx/html nginx
d8ba766b1a8fc49e6dd6d8e3e11e9751a908e3ee2dd5b563e1786664facae26b
root@AveJlanjekar:~# docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
d8ba766b1a8f	nginx	"/docker-entrypoint..."	3 seconds ago	Up 2 seconds	0.0.0.0:8090->80/tcp, [::]:8090->80/tcp	cloudeithx-mynginx
c206bb4c5c09	nginx	"/docker-entrypoint..."	21 minutes ago	Up 21 minutes	80/tcp	nginx-cont2
f248db95f8f0	nginx	"/docker-entrypoint..."	21 minutes ago	Up 21 minutes	80/tcp	nginx-cont1
e559b2bd4f24	cloudeithx-mynginx	"/docker-entrypoint..."	2 hours ago	Up 2 hours	0.0.0.0:8383->80/tcp, [::]:8383->80/tcp	cloudeithx-mynginx-cont
998d04ecba39b	http-server:latest	"httpd-foreground"	3 hours ago	Up 3 hours	0.0.0.0:8181->80/tcp, [::]:8181->80/tcp	webapp
e0ed2dd929db	httpd	"httpd-foreground"	17 hours ago	Up 3 hours	80/tcp	httpd-always

```

root@AveJlanjekar:~# docker exec -it cloudeithx-mynginx /bin/bash
root@d8ba766b1a8f:/# cd /usr/share/nginx
root@d8ba766b1a8f:/usr/share/nginx# ls
html
root@d8ba766b1a8f:/usr/share/nginx# cd html
root@d8ba766b1a8f:/usr/share/nginx/html# ls
50x.html  index.html
root@d8ba766b1a8f:/usr/share/nginx/html# cat index.html
hello cloudeithx , be always team cloudeithx
root@d8ba766b1a8f:/usr/share/nginx/html#

```

```
version: "3.8"

services:
  web:
    image: nginx:latest
    container_name: nginx-web
    ports:
      - "8080:80"
    depends_on:
      - db
    networks:
      - cloudeith-net

  db:
    image: mysql:8.0
    container_name: mysql-db
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: cloudeith
      MYSQL_USER: avejlanjekah
      MYSQL_PASSWORD: cloudeith123
    ports:
      - "3306:3306"
    networks:
      - cloudeith-net

networks:
  cloudeith-net:
    driver: bridge
```

"docker-compose.yml" 29L, 503B

```

root@AvejLanjekar:~# docker compose up -d
WARN[0000] /root/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 14/14
✔db Pulled
✔e3e5d1ac74c1 Pull complete
✔96d30d9fbee8 Pull complete
✔0d74d296605b Pull complete
✔a63160a5eda1 Pull complete
✔7534d1db9f8d Pull complete
✔49ec2dab01d9 Pull complete
✔ab24264a27e9 Pull complete
✔297d04cfe470 Pull complete
✔4c8a3e0d4e4b Pull complete
✔6ef6c7b50a93 Pull complete
✔edf85073f64e Pull complete
✔796812c73292 Download complete
✔af166387641d Download complete
[+] Running 3/3
✔Network root_cloudeithx-net Created
✔Container mysql-db Started
✔Container nginx-web Started
root@AvejLanjekar:~# docker ps
docker: unknown command: docker ps

Run 'docker --help' for more information
root@AvejLanjekar:~# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
55ef75bad5e1   nginx:latest   "/docker-entrypoint..." 2 minutes ago  Up 2 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  nginx-web
da255817babf   mysql:8.0      "docker-entrypoint.s..." 2 minutes ago  Up 2 minutes  0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp  mysql-db

```

```

root@AvejLanjekar:~# docker network inspect root_cloudeithx-net
[
  {
    "Name": "root_cloudeithx-net",
    "Id": "49255a4974d6c3f41394574e5f59a73781cef934bd900dc2637b7cf5dc99fd6a",
    "Created": "2026-08-04T08:24:19.386706357Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "IPRange": "",
          "Gateway": "172.20.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {
      "com.docker.compose.config-hash": "a9e1ccdd4f3a64e4a449f2aa9b40a9429d9e1c9cb1774f5e6e89c8a3673b25e83",
      "com.docker.compose.network": "cloudeithx-net",
      "com.docker.compose.project": "root",
      "com.docker.compose.version": "2.40.3"
    },
    "Containers": [
      {
        "55ef75bad5e1088fb6341d7abc35b6217f7b6c32feb22797558e758cca954d4c": {
          "Name": "nginx-web",
          "EndpointID": "ab9e6b1017576d1341f35258dc9c44b24c2e5fa78beeb8fcae2bd9eaa9b9cd",
          "MacAddress": "4a:f4:ec:7e:cb:cb",
          "IPv4Address": "172.20.0.3/16",
          "IPv6Address": ""
        }
      },
      {
        "da255817babf9f875b6d0ec5c3d46b0d7e8d463e27244b32f7d060092749de50": {
          "Name": "mysql-db",
          "EndpointID": "f0e7123a97d2de1b5cbd598a5e28c3718aefa5908f956a94a9ab32e15d69b27f",
          "MacAddress": "aa:5e:de:1f:bd:be",
          "IPv4Address": "172.20.0.2/16",
          "IPv6Address": ""
        }
      }
    ]
  }
]

```

```

    "com.docker.compose.version": "2.40.3"
  },
  "Containers": {
    "55ef75bad5e1088fb6341d7abc35b6217f7b6c32feb22797558e758cca954d4c": {
      "Name": "nginx-web",
      "EndpointID": "ab9e6b1017576d1341f35258dc9c44b24c2e5fa78beeb8fcae2bd9eaa9b9cd",
      "MacAddress": "4a:f4:ec:7e:cb:cb",
      "IPv4Address": "172.20.0.3/16",
      "IPv6Address": ""
    },
    "da255817babf9f875b6d0ec5c3d46b0d7e8d463e27244b32f7d060092749de50": {
      "Name": "mysql-db",
      "EndpointID": "f0e7123a97d2de1b5cbd598a5e28c3718aefa5908f956a94a9ab32e15d69b27f",
      "MacAddress": "aa:5e:de:1f:bd:be",
      "IPv4Address": "172.20.0.2/16",
      "IPv6Address": ""
    }
  },
  "Status": {
    "IPAM": {
      "Subnets": {
        "172.20.0.0/16": {
          "IPsInUse": 5,
          "DynamicIPsAvailable": 65531
        }
      }
    }
  }
}
]
root@AvejLanjekar:~# |

```

31. Container logging: Run a Docker container that generates log output, and practice using docker logs to view and analyze the logs.

```
root@avejlanjekar:~# docker run -d --name nginx-logs -p 8384:80 nginx:latest
af619cd83ea64e4e1eal549f8f27e4238bbfca02e061f8d5f283acbdce767bdb
root@avejlanjekar:~# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
af619cd83ea6   nginx:latest   "/docker-entrypoint..."   3 seconds ago   Up 2 seconds   0.0.0.0:8384->80/tcp, [::]:8384->80/tcp
Names
nginx-logs
```

```
root@avejlanjekar:~# curl http://localhost:8384
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color:scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

```
root@avejlanjekar:~# docker logs nginx-logs
root@avejlanjekar:~# docker logs nginx-logs
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/08/04 08:33:07 [notice] 1#1: using the "epoll" event method
2026/08/04 08:33:07 [notice] 1#1: nginx/1.24.0
2026/08/04 08:33:07 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/08/04 08:33:07 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/08/04 08:33:07 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026/08/04 08:33:07 [notice] 1#1: start worker processes
2026/08/04 08:33:07 [notice] 1#1: start worker process 29
2026/08/04 08:33:07 [notice] 1#1: start worker process 30
2026/08/04 08:33:07 [notice] 1#1: start worker process 31
2026/08/04 08:33:07 [notice] 1#1: start worker process 32
2026/08/04 08:33:07 [notice] 1#1: start worker process 33
2026/08/04 08:33:07 [notice] 1#1: start worker process 34
2026/08/04 08:33:07 [notice] 1#1: start worker process 35
2026/08/04 08:33:07 [notice] 1#1: start worker process 36
172.17.0.1 - - [04/Aug/2026:08:33:04 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [04/Aug/2026:08:34:04 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:04 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8384/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026/08/04 08:34:04 [error] 29#29: *2 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "loca
localhost:8384", referer: "http://localhost:8384/"
172.17.0.1 - - [04/Aug/2026:08:34:25 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:26 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:26 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:27 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:27 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:27 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:27 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:29 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [04/Aug/2026:08:34:29 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
root@avejlanjekar:~#
```

```
root@avejlanjekar:~# docker logs -t nginx-logs
2026-08-04T08:33:07.234087925Z /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
2026-08-04T08:33:07.234788046Z /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
2026-08-04T08:33:07.234858050Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
2026-08-04T08:33:07.235189151Z 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
2026-08-04T08:33:07.235365332Z 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
2026-08-04T08:33:07.280882712Z /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
2026-08-04T08:33:07.291114172Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
2026-08-04T08:33:07.291889122Z /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
2026-08-04T08:33:07.320803132Z 2026/08/04 08:33:07 [notice] 1#1: using the "epoll" event method
2026-08-04T08:33:07.326140430Z 2026/08/04 08:33:07 [notice] 1#1: nginx/1.24.0
2026-08-04T08:33:07.328075082Z 2026/08/04 08:33:07 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026-08-04T08:33:07.328720802Z 2026/08/04 08:33:07 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026-08-04T08:33:07.328790812Z 2026/08/04 08:33:07 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026-08-04T08:33:07.330887712Z 2026/08/04 08:33:07 [notice] 1#1: start worker processes
2026-08-04T08:33:07.330887712Z 2026/08/04 08:33:07 [notice] 1#1: start worker process 29
2026-08-04T08:33:07.331781732Z 2026/08/04 08:33:07 [notice] 1#1: start worker process 30
2026-08-04T08:33:07.332180732Z 2026/08/04 08:33:07 [notice] 1#1: start worker process 31
2026-08-04T08:33:07.332180732Z 2026/08/04 08:33:07 [notice] 1#1: start worker process 32
2026-08-04T08:33:07.332180732Z 2026/08/04 08:33:07 [notice] 1#1: start worker process 33
2026-08-04T08:33:07.333087762Z 2026/08/04 08:33:07 [notice] 1#1: start worker process 34
2026-08-04T08:33:07.333087762Z 2026/08/04 08:33:07 [notice] 1#1: start worker process 35
2026-08-04T08:33:07.336613552Z 2026/08/04 08:33:07 [notice] 1#1: start worker process 36
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:33:04 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:04 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:04 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8384/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:25 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:26 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:26 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:27 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:27 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:27 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:29 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:33:07.362728082Z 172.17.0.1 - - [04/Aug/2026:08:34:29 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
```

```

root@AvejLanjekar:~# docker logs --tail 5 -t nginx-logs
2026-08-04T08:34:28.074969170Z 172.17.0.1 - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:34:28.624597783Z 172.17.0.1 - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:34:28.808881671Z 172.17.0.1 - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:34:28.953591369Z 172.17.0.1 - [04/Aug/2026:08:34:28 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
2026-08-04T08:34:29.188892891Z 172.17.0.1 - [04/Aug/2026:08:34:29 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36" "-"
root@AvejLanjekar:~#

```

32. Container resource monitoring: Run a Docker container that consumes system resources (e.g., CPU or memory) and practice using docker stats and docker top to monitor the container's resource usage.

```

root@AvejLanjekar:~# docker run -d --name resource-consume ubuntu:latest bash -c "apt-get update && apt-get install -y stress && stress --cpu 2"
4727d8d35dc117bd4b040d650452974f96522275652e5617adb3f430165318cc
root@AvejLanjekar:~# docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
4727d8d35dc1	ubuntu:latest	"bash -c 'apt-get up..."	2 seconds ago	Up 2 seconds		resource-consume
af619cd83ea6	nginx:latest	"/docker-entrypoint..."	10 minutes ago	Up 10 minutes	0.0.0.0:8384->80/tcp, [::]:8384->80/tcp	nginx-logs
55ef75bad5e1	nginx:latest	"/docker-entrypoint..."	18 minutes ago	Up 18 minutes	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	nginx-web
da255817babf	mysql:8.0	"/docker-entrypoint.s..."	18 minutes ago	Up 18 minutes	0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp	mysql-db
ddba766b1a8f	nginx	"/docker-entrypoint..."	26 minutes ago	Up 26 minutes	0.0.0.0:8090->80/tcp, [::]:8090->80/tcp	cloudethix-mynginx
62968bbc5209	nginx	"/docker-entrypoint..."	47 minutes ago	Up 47 minutes	80/tcp	nginx-cont2
fa2a8d95f8f0	nginx	"/docker-entrypoint..."	47 minutes ago	Up 47 minutes	80/tcp	nginx-cont1
a559b2b4df24	cloudethix-nginx	"/docker-entrypoint..."	3 hours ago	Up 3 hours	0.0.0.0:8383->80/tcp, [::]:8383->80/tcp	cloudethix-nginx-co
998d40cba39b	http-server:latest	"httpd-foreground"	3 hours ago	Up 3 hours	0.0.0.0:8181->80/tcp, [::]:8181->80/tcp	webapp
e0ed2dd929db	httpd	"httpd-foreground"	17 hours ago	Up 4 hours	80/tcp	httpd-always

```

root@AvejLanjekar:~# docker stats resource-consume

```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
4727d8d35dc1	resource-consume	200.34%	64.38MiB / 3.725GiB	1.69%	25.6MB / 207kB	35.2MB / 66.8MB	3

```

root@AvejLanjekar:~#
root@AvejLanjekar:~# docker top resource-consume

```

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	71143	71121	0	08:43	?	00:00:00	stress --cpu 2
root	71265	71143	99	08:43	?	00:01:00	stress --cpu 2
root	71266	71143	99	08:43	?	00:01:00	stress --cpu 2

```

root@AvejLanjekar:~#

```

33. Updating a containerized application: Update the source code of a simple containerized application, rebuild the Docker image, and deploy the updated version while minimizing downtime.

```

root@AvejLanjekar:~/CloudEthix-nginx# cat index.html
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthix Assignment</title>
</head>
<body>
  <h1>Welcome to CloudEthix Nginx Docker Container</h1>
</body>
</html>
root@AvejLanjekar:~/CloudEthix-nginx#

```

```
root@bawelanjekar:~/CloudEthix-nginx# via index.html
root@bawelanjekar:~/CloudEthix-nginx# cat index.html
<DOCTYPE html>
<html>
<head>
<title>CloudEthix Assignment</title>
</head>
<body>
<h1>Welcome to CloudEthix Nginx Docker Container</h1>
<h2> updated successfully</h2>
</body>
</html>
root@bawelanjekar:~/CloudEthix-nginx# docker build cloudthix-imp
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

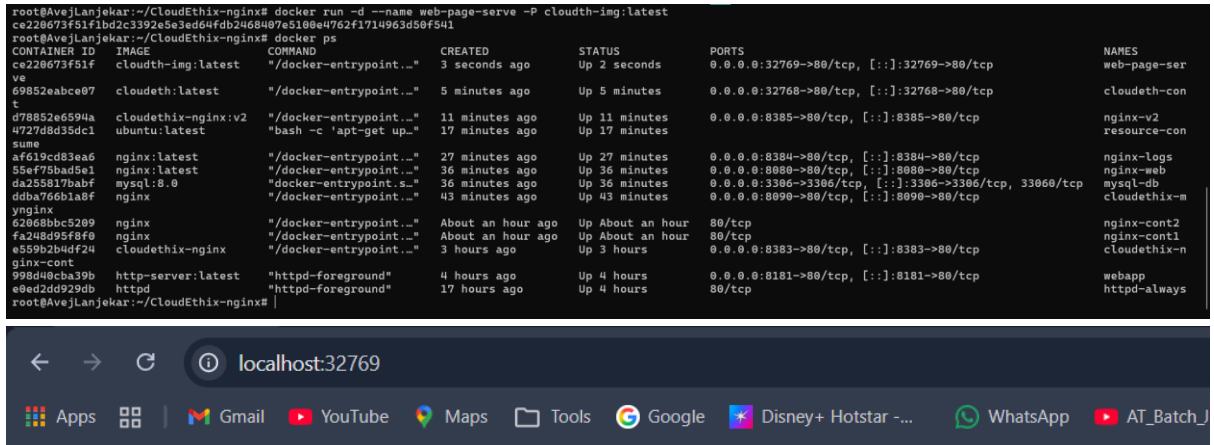
docker: 'docker build' requires 1 argument

Usage: docker build [OPTIONS] PATH | URL | -

Run 'docker build --help' for more information
root@bawelanjekar:~/CloudEthix-nginx# docker build -t cloudthix-imp .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

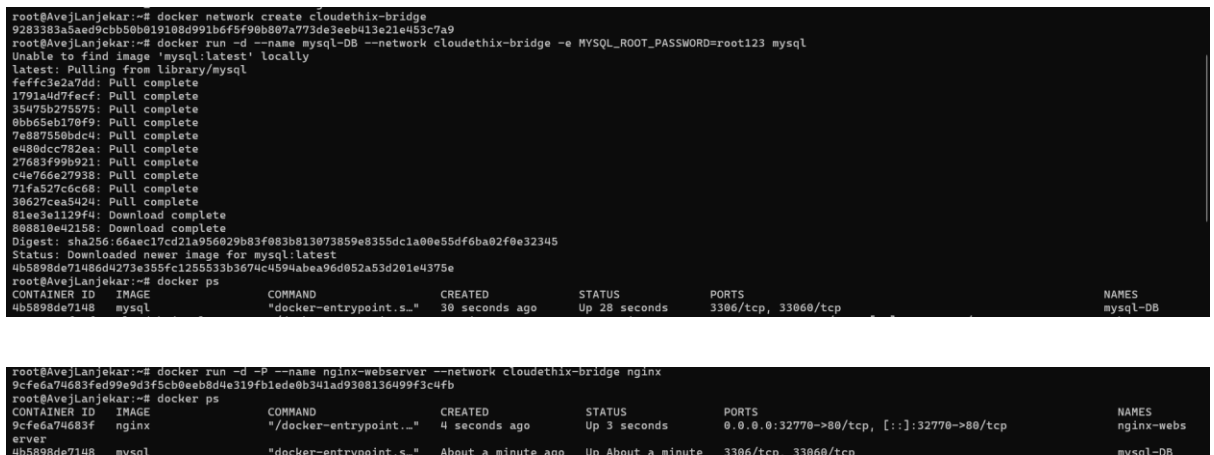
Sending build context to Docker daemon 168.3MB
Step 1/3 : FROM nginx:latest
--> 5a889c9c5479
Step 2/3 : COPY index.html /usr/share/nginx/html/index.html
--> 32ab33e2811d
Step 3/3 : EXPOSE 80
--> Running in 01e455f36356
--> Removed intermediate container 01e455f36356
--> d6872226cedc
Successfully built d6872226cedc
Successfully tagged cloudthix-imp:latest
```

Welcome to CloudEthix Nginx Docker Container



Welcome to CloudEthix Nginx Docker Container updated successfully

34. Bridge network: Create two Docker containers using different images, such as NGINX and MySQL. Connect them using a user-defined bridge network and ensure they can communicate with each other.



```

root@AvejLanJekar:~# docker network inspect cloudethix-bridge
[
  {
    "Name": "cloudethix-bridge",
    "Id": "9283383a5aed9cbb58b619108d991b6f5f90b807a773de3eeb413e21e453c7a9",
    "Created": "2026-08-04T09:37:14.097046579Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.21.0.0/16",
          "IPRange": "",
          "Gateway": "172.21.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {
      "4b5898de71486dd4273e355fcl255533b3674c4594abea96d052a53d201e4375e": {
        "Name": "mysql-DB",
        "EndpointID": "c77587276cd3059ed942266ae8f37556d765d6c386309ed12efed2d036c8e32",
        "MacAddress": "8e:e6:50:1f:f9:d3",
        "IPv4Address": "172.21.0.2/16",
        "IPv6Address": ""
      },
      "9cf6ea74683fed99e9d3f5cb8eeb8d4e319fb1ede0b341ad9308136499f3c4fb": {
        "Name": "nginx-webserver",
        "EndpointID": "2b05c7c63e31934fd05d5159fad2bd25f42f6091dd53b1487b6204a841f0e800",
        "MacAddress": "06:b7:8a:c1:a6:0e",
        "IPv4Address": "172.21.0.3/16",
        "IPv6Address": ""
      }
    },
    "Status": {
      "IPAM": {
        "Subnets": [
          {
            "172.21.0.0/16": {
              "IPsInUse": 5,
              "DynamicIPsAvailable": 65531
            }
          }
        ]
      }
    }
  }
]
root@AvejLanJekar:~#

```

```

root@AvejLanJekar:~# docker exec -it nginx-webserver /bin/bash
root@9cf6ea74683f:/# ping -c 3 mysql-DB
PING mysql-DB (172.21.0.2) 56(84) bytes of data.
64 bytes from mysql-DB.cloudethix-bridge (172.21.0.2): icmp_seq=1 ttl=64 time=3.23 ms
64 bytes from mysql-DB.cloudethix-bridge (172.21.0.2): icmp_seq=2 ttl=64 time=0.083 ms
64 bytes from mysql-DB.cloudethix-bridge (172.21.0.2): icmp_seq=3 ttl=64 time=0.088 ms

--- mysql-DB ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2031ms
rtt min/avg/max/mdev = 0.083/1.134/3.231/1.482 ms
root@9cf6ea74683f:/# exit

```

35. Port mapping: Run a containerized web server (e.g., NGINX or Apache) and map its default port (80 or 443) to a custom port on the host machine. Verify that the web server is accessible through the custom port on the host.

```

root@AvejLanJekar:~# docker run -d --name nginx-port -p 9090:80 nginx
5a77f5d86e2a4937ef8e524ab8e2a3dbcded0b78238aa93fb3ee0ca9b0fd5a
root@AvejLanJekar:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
5a77f5d86e2a   nginx    "/docker-entrypoint. ..." 4 seconds ago   Up 3 seconds   0.0.0.0:9090->80/tcp, [::]:9090->80/tcp
9cf6ea74683f   nginx    "/docker-entrypoint. ..." 18 minutes ago   Up 18 minutes   0.0.0.0:32778->80/tcp, [::]:32778->80/tcp
4b5898de7109   mysql    "docker-entrypoint.sh ..." 11 minutes ago   Up 11 minutes   3306/tcp, 33060/tcp

```

localhost:9090

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
 To engage with the community please visit community.nginx.org.
 For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

```

root@AvejLanjekar:~# curl http://localhost:9090
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@AvejLanjekar:~# |

```

```

root@AvejLanjekar:~#
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker port nginx-port
80/tcp -> 0.0.0.0:9090
80/tcp -> [::]:9090
root@AvejLanjekar:~# |

```

36. Host networking: Run a container with the host networking mode and compare its network configuration with that of the host machine. Explain the advantages and disadvantages of using host networking for containers.

```

root@AvejLanjekar:~# hostname -I
127.0.1.1
root@AvejLanjekar:~# hostname -I
172.23.31.132 17.0.0.1 172.17.0.1 172.18.0.1 172.19.0.1 172.20.0.1 172.21.0.1
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker run -d --name nginx-host --network host nginx
99ce4e073ab69f3b8aada6de304e48b48caede73a51a870f40978c8658e237d
root@AvejLanjekar:~# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
99ce4e073ab6   nginx     "/docker-entrypoint..." 8 seconds ago  Up 7 seconds  0.0.0.0:9090->80/tcp, [::]:9090->80/tcp
54d7fe5d06a2   nginx     "/docker-entrypoint..." 6 minutes ago  Up 6 minutes  0.0.0.0:32778->80/tcp, [::]:32778->80/tcp
4b5898de71d8   mysql     "docker-entrypoint.s..." 17 minutes ago Up 17 minutes  3306/tcp, 33060/tcp
root@AvejLanjekar:~# docker exec -it nginx-host /bin/bash
root@AvejLanjekar:/# hostname -I
172.23.31.132 17.0.0.1 172.17.0.1 172.18.0.1 172.19.0.1 172.20.0.1 172.21.0.1
root@AvejLanjekar:/#
root@AvejLanjekar:/#
root@AvejLanjekar:/# curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@AvejLanjekar:/# |

```


Advantages:-

Better Performance :- The container uses the host's network directly, so communication is faster.

No Port Mapping Needed:- You don't need to use the -p option because the container uses the host's ports directly.

Easy Access:- Applications running in the container are directly available through the host's IP address.

Disadvantages:-

Port Conflicts:- If the host is already using a port (for example, port 80), the container cannot use the same port.

Not Ideal for Multiple Same Services:- You cannot run two web servers on the same host network if both need the same port (such as port 80).

37. Named volume: Launch a containerized database (e.g., MySQL or PostgreSQL) using a named volume to store its data. Stop and remove the container, then recreate it using the same named volume. Verify that the data persists across container recreations

```
root@AvejLanjekar:~# docker volume create mysql-data
mysql-data
root@AvejLanjekar:~# docker volume ls
DRIVER      VOLUME NAME
local      b1333a942ef0bd3545f2e3be7ecaa4000da151312553b53e1737a615ed5a39d2
local      cloudethix-volume
local      da12b6c72f39118e9f1bc684b964c312128d62a9669d2a2fe13207faec3befe3
local      data-vol
local      data-volume
local      docker-compose_nginx-data
local      my-volume
local      mysql-data
local      nginx-volume
local      shared-data

root@AvejLanjekar:~# docker run -d --name mysql-server -e MYSQL_ROOT_PASSWORD=root1234 -p 3307:3306 -v mysql-data:/var/lib/mysql mysql
1e14cfc4e868c70b8e6906eb17c5beb4c70ed2d0b9e9a45845a970640e7e8cfd
root@AvejLanjekar:~# docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
1e14cfc4e868   mysql    "docker-entrypoint.s..." 8 seconds ago Up 7 seconds  3306/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp  mysql-server

root@AvejLanjekar:~#
root@AvejLanjekar:~# docker exec -it mysql-server mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 26.7.0 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE cloudethix;
ERROR 1049 (42900): Unknown database 'cloudethix'
mysql> create database cloudethix;
Query OK, 1 row affected (0.012 sec)

mysql> USE cloudethix;
Database changed
mysql> CREATE TABLE student(
-> id INT PRIMARY KEY,
-> name VARCHAR(50)
-> );
Query OK, 0 rows affected (0.029 sec)

mysql> INSERT INTO student VALUES
-> (1,'Avej');
Query OK, 1 row affected (0.023 sec)

mysql> select * from students;
ERROR 1146 (42S02): Table 'cloudethix.students' doesn't exist
mysql> select * from student;
+----+-----+
| id | name |
+----+-----+
| 1  | Avej |
+----+-----+
1 row in set (0.000 sec)

mysql> |
```

```

root@AvejLanjekar:~# docker stop mysql-server
mysql-server
root@AvejLanjekar:~# docker rm mysql-server
mysql-server
root@AvejLanjekar:~#
root@AvejLanjekar:~#

```

```

root@AvejLanjekar:~# docker run -d --name mysql-server1 -e MYSQL_ROOT_PASSWORD=root1234 -p 3307:3306 -v mysql-data:/var/lib/mysql mysql
cb3893cff8cf6db0130da88619b40b4b8a288ba3ab7f0665f8e8efecb4be6af
root@AvejLanjekar:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
cb3893cff8cf   mysql    "docker-entrypoint.s..." 4 seconds ago  Up 3 seconds  33060/tcp, 0.0.0.0:3307->3306/tcp  mysql-server1

```

```

root@AvejLanjekar:~# docker exec -it mysql-server1 mysql -uroot -p
Enter password:

```

```

Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 26.7.0 MySQL Community Server - GPL

```

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its affiliates. Other names may be trademarks of their respective owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```

mysql> select * from student;
ERROR 1046 (3D000): No database selected
mysql> show databases;

```

```

+-----+
| Database |
+-----+
| cloudethix |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.011 sec)

```

```

mysql> use cloudethix;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

```

```

Database changed
mysql> select * from student;
+----+-----+
| id | name |
+----+-----+
| 1 | Avej |
+----+-----+
1 row in set (0.000 sec)

```

```

mysql> |

```

38. Bind mount: Create a simple web application with a local directory containing HTML, CSS, and JavaScript files. Run a web server container (e.g., NGINX or Apache) and bind mount the local directory to the container's webroot. Verify that the web application is accessible and update the local files to see if changes are reflected in the container

```
root@AvejLanjekar:~/bind-mount-app# vim index.html
root@AvejLanjekar:~/bind-mount-app# vim style.css
root@AvejLanjekar:~/bind-mount-app# vim script.js
root@AvejLanjekar:~/bind-mount-app# ls
index.html  script.js  style.css
root@AvejLanjekar:~/bind-mount-app# cat index.html
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Hello, Team CloudEthiX!</h1>
  <button onClick="showMessage()">Click Me</button>

  <script src="script.js"></script>
</body>
</html>
root@AvejLanjekar:~/bind-mount-app# cat style.css
cat: style.css: No such file or directory
root@AvejLanjekar:~/bind-mount-app# cat style.css
body{
  font-family: Arial;
  background: lightblue;
  text-align:center;
  margin-top:80px;
}

h1{
  color:darkblue;
}

button{
  padding:10px 20px;
  font-size:18px;
}
root@AvejLanjekar:~/bind-mount-app# cat script.js
function showMessage(){
  alert("Welcome to Team CloudEthiX!");
}
root@AvejLanjekar:~/bind-mount-app#
```

```
root@AvejLanjekar:~/bind-mount-app# docker run -d --name bind-nginx -P -v $(pwd):/usr/share/nginx/html nginx
5d8393f4cf8bb12cbb219428963ec78a89df3cdd1de31a4242e7abe7983fa97c
root@AvejLanjekar:~/bind-mount-app# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
5d8393f4cf8bb  nginx         "/docker-entrypoint..." 4 seconds ago  Up 4 seconds  0.0.0.0:32771->80/tcp, [::]:32771->80/tcp  bind-nginx
cb3893cfc8fcf  mysql        "/docker-entrypoint.s..." 6 minutes ago  Up 6 minutes  3306/tcp, 0.0.0.0:3387->3306/tcp, [::]:3307->3306/tcp  mysql-server1
99ce4e073ab6   nginx        "/docker-entrypoint..." 34 minutes ago  Up 34 minutes  0.0.0.0:9090->80/tcp, [::]:9090->80/tcp  nginx-host
54477fe5d06a2  nginx        "/docker-entrypoint..." 48 minutes ago  Up 48 minutes  0.0.0.0:32770->80/tcp, [::]:32770->80/tcp  nginx-port
9cfa6a74683f   nginx        "/docker-entrypoint..." 58 minutes ago  Up 58 minutes  3306/tcp, 3306/tcp                  nginx-webserver
4b5898de7148   mysql        "/docker-entrypoint.s..." 51 minutes ago  Up 51 minutes  0.0.0.0:32769->80/tcp, [::]:32769->80/tcp  mysql-db
ce228673f5f1   cloudeth:img:latest "/docker-entrypoint..." 2 hours ago    Up 2 hours    0.0.0.0:32768->80/tcp, [::]:32768->80/tcp  web-page-serve
69852eabce07   cloudeth:latest  "/docker-entrypoint..." 2 hours ago    Up 2 hours    0.0.0.0:8385->80/tcp, [::]:8385->80/tcp  cloudeth-cont
d78852ee594a   cloudeth:nginx:v2 "/docker-entrypoint..." 2 hours ago    Up 2 hours    0.0.0.0:8384->80/tcp, [::]:8384->80/tcp  nginx-v2
4727d083d6c1   ubuntu:latest  "bash -c 'apt-get up..." 2 hours ago    Up 2 hours    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  resource-consume
af619cd83ea6   nginx:latest   "/docker-entrypoint..." 2 hours ago    Up 2 hours    0.0.0.0:8386->80/tcp, [::]:8386->80/tcp  nginx-logs
55ef75bad5e1   nginx:latest   "/docker-entrypoint..." 2 hours ago    Up 2 hours    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  nginx-web
da255817babf   mysql:8.0      "/docker-entrypoint.s..." 2 hours ago    Up 2 hours    0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 3306/tcp  mysql-db
ddb8a7661a8f   nginx          "/docker-entrypoint..." 2 hours ago    Up 2 hours    0.0.0.0:8090->80/tcp, [::]:8090->80/tcp  cloudethix-mynginx
62968b0bc5209   nginx         "/docker-entrypoint..." 3 hours ago    Up 3 hours    80/tcp  nginx-cont2
fa248d95f8f0   nginx         "/docker-entrypoint..." 3 hours ago    Up 3 hours    80/tcp  nginx-cont1
e559b2b4df24   cloudethix-nginx "/docker-entrypoint..." 4 hours ago    Up 4 hours    0.0.0.0:8383->80/tcp, [::]:8383->80/tcp  cloudethix-nginx-cont
998d48cba39b   http-server:latest "httdp-foreground"      5 hours ago    Up 5 hours    0.0.0.0:8181->80/tcp, [::]:8181->80/tcp  webapp
e8ed2d0929db   httdp        "httdp-foreground"      19 hours ago   Up 6 hours    80/tcp  httdp-always
```

Hello, Team CloudEthiX!

Click Me

39. Volume backups: Using a container with a named volume, create a backup of the volume's data by either exporting it as a tarball or copying the data to a host directory. Restore the data to a new container and verify that the restoration was successful.

Step 1: Create a Named Volume

```
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker volume create backup-volume
backup-volume
root@AvejLanjekar:~# docker volume ls
DRIVER    VOLUME NAME
local     b1333a942ef0bd3545f2e3be7ecaa400da151312553b53e1737a615ed5a39d2
local     backup-volume
local     cloudeithix-volume
local     dal2b6c72f39118e9f1bc684b964c312128d62a9669d2a2fe13207faec3befe3
local     data-vol
local     data-volume
local     docker-compose_nginx-data
local     my-volume
local     mysql-data
local     nginx-volume
local     shared-data
root@AvejLanjekar:~# |
```

Step 2: Run an NGINX Container Using the Volume. Enter the container & create file

```
root@AvejLanjekar:~# docker run -d --name nginx-backup -v backup-volume:/usr/share/nginx/html -P nginx
3301358aba091aab7edcb2983b12d5d8966e385746f17fe28fe18a6529ba2b24
root@AvejLanjekar:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
3301358aba09   nginx    "/docker-entrypoint..." 11 seconds ago Up 11 seconds 0.0.0.0:32772->80/tcp, [::]:32772->80/tcp
5d8393f4cf8b   nginx    "/docker-entrypoint..." 6 minutes ago  Up 6 minutes  0.0.0.0:32771->80/tcp, [::]:32771->80/tcp
99cedae973ab6  nginx    "/docker-entrypoint..." 48 minutes ago Up 48 minutes  33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
54477fe5d06a2  nginx    "/docker-entrypoint..." 46 minutes ago Up 46 minutes  0.0.0.0:9090->80/tcp, [::]:9090->80/tcp
9cfe6a74683f   nginx    "/docker-entrypoint..." 56 minutes ago Up 56 minutes  0.0.0.0:32770->80/tcp, [::]:32770->80/tcp
4b5898de7148   mysql    "/docker-entrypoint..." 58 minutes ago Up 58 minutes  3306/tcp, 33060/tcp
e22a673f51f    cloudeithix:latest 2 hours ago    Up 2 hours    0.0.0.0:32769->80/tcp, [::]:32769->80/tcp
69852eabc0e7   cloudeithix:latest 2 hours ago    Up 2 hours    0.0.0.0:32768->80/tcp, [::]:32768->80/tcp
478852e6594a   cloudeithix-nginx:v2 2 hours ago    Up 2 hours    0.0.0.0:8385->80/tcp, [::]:8385->80/tcp
4727d8d35dc1   ubuntu:latest      2 hours ago    Up 2 hours    0.0.0.0:8384->80/tcp, [::]:8384->80/tcp
af619cd083ea6  nginx:latest        2 hours ago    Up 2 hours    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
95e775bada5e1  nginx:latest        2 hours ago    Up 2 hours    0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp
4a255817babf    mysql:8.0           2 hours ago    Up 2 hours    0.0.0.0:8090->80/tcp, [::]:8090->80/tcp
d4ba766b1a8f    nginx               2 hours ago    Up 2 hours    80/tcp
62068b5c5209    nginx               3 hours ago    Up 3 hours    80/tcp
fa248d95f8f0    nginx               3 hours ago    Up 3 hours    0.0.0.0:8383->80/tcp, [::]:8383->80/tcp
e559b2b4d424    cloudeithix-nginx   5 hours ago    Up 5 hours    0.0.0.0:8181->80/tcp, [::]:8181->80/tcp
998d40c3a39b    http-server:latest  5 hours ago    Up 5 hours    80/tcp
e8ed2dd929db    httpd              19 hours ago   Up 6 hours    80/tcp
root@AvejLanjekar:~#
root@AvejLanjekar:~# docker exec -it nginx-backup /bin/bash
root@3301358aba09:/# echo "<h1>Docker Volume Backup</h1>" > /usr/share/nginx/html/index.html
root@3301358aba09:/# exit
exit
```

Step 3: Create a Backup Directory on the Host

```
root@AvejLanjekar:~#
root@AvejLanjekar:~# mkdir ~/volume-backup
root@AvejLanjekar:~# ls
Built-Dockerfile  bind-mount-app  docker-compose  docker-compose.yml  file.txt  index.html  test1  volume-backup
CloudEithix-nginx  compose-demo    docker-compose-lab  docker-project      httpd     nginx-container.tar  ubuntu-nginx
root@AvejLanjekar:~# cd ~/volume-backup
root@AvejLanjekar:~/volume-backup# pwd
/root/volume-backup
root@AvejLanjekar:~/volume-backup#
root@AvejLanjekar:~/volume-backup#
root@AvejLanjekar:~/volume-backup#
root@AvejLanjekar:~/volume-backup#
root@AvejLanjekar:~/volume-backup#
```

Step 4: Backup the Volume as a Tar File. Create a New Volume , Restore the Backup, Run a New NGINX Container from restored volume

```
root@AvejLanjekar:~/volume-backup# docker run --rm -v backup-volume:/volume -v ~/volume-backup:/backup ubuntu tar czf /backup/backup.tar.gz -C /volume .
root@AvejLanjekar:~/volume-backup# ls -l /volume-backup
-rw-r--r-- 1 root root 1024000 2024-08-28 10:10 backup.tar.gz
root@AvejLanjekar:~/volume-backup# docker volume create restored-volume
restored-volume
root@AvejLanjekar:~/volume-backup# docker run --rm -v restored-volume:/restore -v ~/volume-backup:/backup ubuntu bash -c "cd /restore && tar xzf /backup/backup.tar.gz"
root@AvejLanjekar:~/volume-backup#
root@AvejLanjekar:~/volume-backup# docker run -d --name nginx-restored -v restored-volume:/usr/share/nginx/html -P nginx
d82688fab52545c9a2c5d20858b5e7640c1c2d
root@AvejLanjekar:~/volume-backup# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
d82688fab525   nginx    "/docker-entrypoint..." 3 seconds ago  Up 3 seconds  0.0.0.0:32773->80/tcp, [::]:32773->80/tcp
3301358aba09   nginx    "/docker-entrypoint..." 8 minutes ago  Up 8 minutes  0.0.0.0:32772->80/tcp, [::]:32772->80/tcp
5d8393f4cf8b   nginx    "/docker-entrypoint..." 15 minutes ago Up 15 minutes  0.0.0.0:32771->80/tcp, [::]:32771->80/tcp
99cedae973ab6  mysql    "/docker-entrypoint..." 21 minutes ago Up 21 minutes  33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
54477fe5d06a2  nginx    "/docker-entrypoint..." 55 minutes ago Up 55 minutes  0.0.0.0:9090->80/tcp, [::]:9090->80/tcp
9cfe6a74683f   nginx    "/docker-entrypoint..." About an hour ago Up About an hour  0.0.0.0:32770->80/tcp, [::]:32770->80/tcp
4b5898de7148   mysql    "/docker-entrypoint..." About an hour ago Up About an hour  3306/tcp, 33060/tcp
e22a673f51f    cloudeithix:latest 2 hours ago    Up 2 hours    0.0.0.0:32769->80/tcp, [::]:32769->80/tcp
69852eabc0e7   cloudeithix:latest 2 hours ago    Up 2 hours    0.0.0.0:32768->80/tcp, [::]:32768->80/tcp
478852e6594a   cloudeithix-nginx:v2 2 hours ago    Up 2 hours    0.0.0.0:8385->80/tcp, [::]:8385->80/tcp
4727d8d35dc1   ubuntu:latest      2 hours ago    Up 2 hours    0.0.0.0:8384->80/tcp, [::]:8384->80/tcp
af619cd083ea6  nginx:latest        2 hours ago    Up 2 hours    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
95e775bada5e1  nginx:latest        2 hours ago    Up 2 hours    0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp
4a255817babf    mysql:8.0           2 hours ago    Up 2 hours    0.0.0.0:8090->80/tcp, [::]:8090->80/tcp
d4ba766b1a8f    nginx               3 hours ago    Up 3 hours    80/tcp
62068b5c5209    nginx               3 hours ago    Up 3 hours    80/tcp
fa248d95f8f0    nginx               3 hours ago    Up 3 hours    0.0.0.0:8383->80/tcp, [::]:8383->80/tcp
e559b2b4d424    cloudeithix-nginx   5 hours ago    Up 5 hours    0.0.0.0:8181->80/tcp, [::]:8181->80/tcp
998d40c3a39b    http-server:latest  5 hours ago    Up 5 hours    80/tcp
e8ed2dd929db    httpd              19 hours ago   Up 6 hours    80/tcp
root@AvejLanjekar:~/volume-backup#
```

Step 5:- Verify the Restoration:-

```
root@AvejLanjekar:~/volume-backup# curl http://localhost:32773
<h1>Docker Volume Backup</h1>
root@AvejLanjekar:~/volume-backup# |
```

40. Docker Compose networking: Create a Docker Compose file that defines a multi-container application with a frontend, backend, and database. Set up custom networks and volumes for the services and ensure that they can communicate with each other and store data persistently

```
root@AvejLanjekar:~/docker-compose-network# vim docker-compose.yml
root@AvejLanjekar:~/docker-compose-network#
```

```
version: "3.8"
services:
  frontend:
    image: nginx:latest
    container_name: frontend
    ports:
      - "8087:80"
    depends_on:
      - backend
    networks:
      - frontend-network
  backend:
    image: httpd:latest
    container_name: backend
    ports:
      - "8088:80"
    depends_on:
      - database
    networks:
      - frontend-network
      - backend-network
  database:
    image: mysql:8.0
    container_name: database
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: cloudethix
      MYSQL_USER: clouduser
      MYSQL_PASSWORD: cloud123
    volumes:
      - mysql-data:/var/lib/mysql
    ports:
      - "3306:3306"
    networks:
      - backend-network
volumes:
  mysql-data:
networks:
  frontend-network:
    driver: bridge
  backend-network:
    driver: bridge
"docker-compose.yml" 55L, 823B
```

```

root@AveJlanjekar:~/docker-compose-network#
root@AveJlanjekar:~/docker-compose-network# docker compose up -d
WARN[0000] /root/.docker-compose-network/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[*] Running 6/6
✓Network docker-compose-network_frontend-network Created
✓Network docker-compose-network_backend-network Created
✓Volume docker-compose-network_mysql-data Created
✓Container database Started
✓Container backend Started
✓Container frontend Started
root@AveJlanjekar:~/docker-compose-network# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                               NAMES
1bea8e76d048   nginx:latest   "/docker-entrypoint..." 14 seconds ago   Up 12 seconds   0.0.0.0:8087->80/tcp, [::]:8087->80/tcp   frontend
659d8bcbcff6   httpd:latest   "httpd-foreground"        14 seconds ago   Up 13 seconds   0.0.0.0:8088->80/tcp, [::]:8088->80/tcp   backend
32e5ab86776d   mysql:8.0      "docker-entrypoint.s..." 14 seconds ago   Up 13 seconds   33060/tcp, 0.0.0.0:3309->3306/tcp, [::]:3309->3306/tcp   database

root@AveJlanjekar:~/docker-compose-network# docker network ls
NETWORK ID     NAME                        DRIVER  SCOPE
6cc8c4715896   alpine-net                 bridge  local
d8db692f7366   bridge                     bridge  local
9283383a5aed   cloudeithix-bridge         bridge  local
d2a493523b34   docker-compose-network_backend-network  bridge  local
597312a87cf4   docker-compose-network_frontend-network  bridge  local
18f0646935f9   docker-compose_default     bridge  local
ab3dc90848a6   docker-lab                 bridge  local
fd609fea3662   host                       host    local
732a8745d38e   none                       null    local
49255a4974d6   root_cloudeithix-net       bridge  local
root@AveJlanjekar:~/docker-compose-network# |

```

```

root@AveJlanjekar:~/docker-compose-network# docker network inspect docker-compose-network_frontend-network
[
  {
    "Name": "docker-compose-network_frontend-network",
    "Id": "597312a87cf49d67723784c9322c691e504f13011963c19f8e9a3c7a97b97b5",
    "Created": "2026-08-04T11:02:47.899540215Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.22.0.0/16",
          "IPRange": "",
          "Gateway": "172.22.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {
      "com.docker.compose.config-hash": "79bbf75cb262f952b30f6cb55ed5903e09a6c89874122372eabb0812179eecc9",
      "com.docker.compose.network": "frontend-network",
      "com.docker.compose.project": "docker-compose-network",
      "com.docker.compose.version": "2.40.3"
    },
    "Containers": {
      "1bea8e76d048e0879d90893c96948ae3028882872fb1e60890ceae3cd8d2601": {
        "Name": "frontend",
        "EndpointID": "abacddb94d60cc100fe958b07a9de4b141c0431ed135e94cd3bcbfacc1ef4a65",
        "MacAddress": "fe:c7:89:cb:58:b4",
        "IPv4Address": "172.22.0.3/16",
        "IPv6Address": ""
      },
      "659d8bcbcff61ddb1ac4612867c57726cd83e5424144f12c07a85435e3dc31cd": {
        "Name": "backend",
        "EndpointID": "479c1c2b3cfbba2231730b14f8fd5457908501d9289a28913b5fe243b31ce2c4",
        "MacAddress": "f6:25:50:09:47:05",
        "IPv4Address": "172.22.0.2/16",
        "IPv6Address": ""
      }
    }
  }
]

```

```

root@AveJlanjekar:~/docker-compose-network# docker network inspect docker-compose-network_backend-network
[
  {
    "Name": "docker-compose-network_backend-network",
    "Id": "d2a493523b34cdd43ed45735d0a6a9a344a51a9fbc8df4c8054c0ff6cfa7fe",
    "Created": "2026-08-04T11:02:48.070972104Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.24.0.0/16",
          "IPRange": "",
          "Gateway": "172.24.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {
      "com.docker.compose.config-hash": "dba82a7b7911be40297d70a46c2199103bedeb6c533056dc19c34a5c41beb127",
      "com.docker.compose.network": "backend-network",
      "com.docker.compose.project": "docker-compose-network",
      "com.docker.compose.version": "2.40.3"
    },
    "Containers": {
      "32e5ab86770d1dc532c95064761a059e31715a48d592a803ccd8cbaecb90f94": {
        "Name": "database",
        "EndpointID": "82e22cd08021346230eba6156f0b2f725f1b11f56452cc39e279426d2307a750",
        "MacAddress": "d2:ce:22:aa:07:02",
        "IPv4Address": "172.24.0.2/16",
        "IPv6Address": ""
      },
      "659d8bcbcff61ddb1ac4612867c57726cd83e5424144f12c07a85435e3dc31cd": {
        "Name": "backend",
        "EndpointID": "0f5d566892fb1d6cd2cd826c8a86c308365624ef4c67d3f9a722543b8f9e81e",
        "MacAddress": "4e:84:17:89:4d:6f",
        "IPv4Address": "172.24.0.3/16",
        "IPv6Address": ""
      }
    }
  }
]

```

```

root@AvejLanjekar:~/docker-compose-network# docker volume ls
DRIVER          VOLUME NAME
local           b1333a942ef0bd3545f2e3be7ecaa4000da151312553b53e1737a615ed5a39d2
local           backup-volume
local           cloudethix-volume
local           dal2b6c72f39118e9f1bc684b964c312128d62a9669d2a2fe13207faec3befe3
local           data-vol
local           data-volume
local           docker-compose-network_mysql-data
local           docker-compose_nginx-data
local           my-volume
local           mysql-data
local           nginx-volume
local           restored-volume
local           shared-data
root@AvejLanjekar:~/docker-compose-network# docker volume inspect docker-compose-network_mysql-data
[
  {
    "CreatedAt": "2026-08-04T11:02:48Z",
    "Driver": "local",
    "Labels": {
      "com.docker.compose.config-hash": "7e5d613ebe7d03c2bb944215a49e4383621abccad539a3d4a65b587d4412aa9b",
      "com.docker.compose.project": "docker-compose-network",
      "com.docker.compose.version": "2.40.3",
      "com.docker.compose.volume": "mysql-data"
    },
    "Mountpoint": "/var/lib/docker/volumes/docker-compose-network_mysql-data/_data",
    "Name": "docker-compose-network_mysql-data",
    "Options": null,
    "Scope": "local"
  }
]

```

```

root@AvejLanjekar:~/docker-compose-network# docker exec -it backend bash
root@659ddbc6cf6:/usr/local/apache2# apt update
Hit:1 http://deb.debian.org/debian trixie InRelease
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9673 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5912 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [231 kB]
Fetched 9999 kB in 6s (1618 kB/s)
1 package can be upgraded. Run 'apt list --upgradable' to see it.
root@659ddbc6cf6:/usr/local/apache2# apt install -y iputils-ping
Installing:
  iputils-ping

Installing dependencies:
  linux-sysctl-defaults

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 1
  Download size: 57.0 kB
  Space needed: 211 kB / 1019 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 iputils-ping amd64 3:20240905-3 [51.2 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 linux-sysctl-defaults all 4.12.1 [5724 B]
Fetched 57.0 kB in 0s (372 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79, <STDIN> line 2.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/Readline.pm in @INC (you may need to install the Term::Readline module) (@INC entries checked: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.40.1 /usr/local/share/perl/5.40.1 /usr/lib/x86_64-linux-gnu/perl5/5.40 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8, <STDIN> line 2.)
debconf: falling back to frontend: Teletype
Selecting previously unselected package iputils-ping.
(Reading database ... 5882 files and directories currently installed.)
Preparing to unpack .../iputils-ping_3:20240905-3_amd64.deb ...
Unpacking iputils-ping (3:20240905-3) ...
Selecting previously unselected package linux-sysctl-defaults.
Preparing to unpack .../linux-sysctl-defaults_4.12.1_all.deb ...
Unpacking linux-sysctl-defaults (4.12.1) ...
Setting up linux-sysctl-defaults (4.12.1) ...
Setting up iputils-ping (3:20240905-3) ...
root@659ddbc6cf6:/usr/local/apache2# ping -c 3 database
PING database (172.24.0.2) 56(84) bytes of data:
64 bytes from database.docker-compose-network_backend-network (172.24.0.2): icmp_seq=1 ttl=64 time=3.42 ms
64 bytes from database.docker-compose-network_backend-network (172.24.0.2): icmp_seq=2 ttl=64 time=0.117 ms
64 bytes from database.docker-compose-network_backend-network (172.24.0.2): icmp_seq=3 ttl=64 time=0.100 ms

--- database ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2030ms
rtt min/avg/max/mdev = 0.100/1.211/3.416/1.559 ms
root@659ddbc6cf6:/usr/local/apache2# exit
exit

```

```

root@AvejLanjekar:~/docker-compose-network# docker exec -it database mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| cloudethix |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.04 sec)

mysql> exit
Bye
root@AvejLanjekar:~/docker-compose-network#
root@AvejLanjekar:~/docker-compose-network#
root@AvejLanjekar:~/docker-compose-network#
root@AvejLanjekar:~/docker-compose-network# docker compose down
WARN[0001] /root/docker-compose-network/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 5/5
  ✓ Container frontend      Removed
  ✓ Container backend       Removed
  ✓ Container database       Removed
  ✓ Network docker-compose-network_frontend-network  Removed
  ✓ Network docker-compose-network_backend-network   Removed
root@AvejLanjekar:~/docker-compose-network#

```